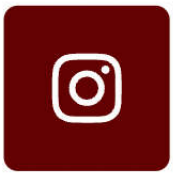
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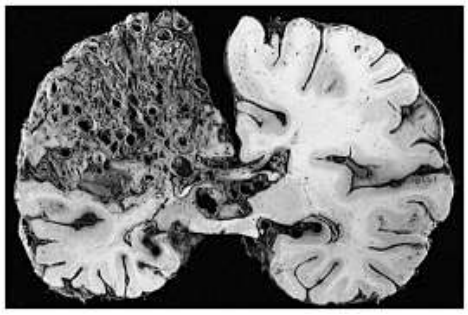
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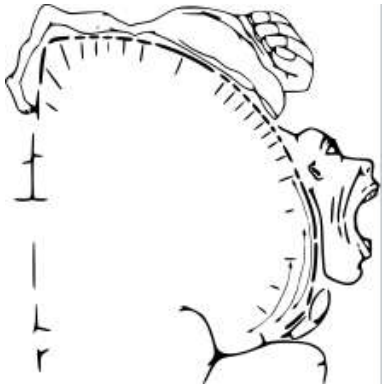
HY Neuroanatomy

The point of this review is not to be a 600-page low-yield neuroanatomical atlas that stokes your OCD neuro professor's penchant for superfluous detail. The aim is to drive home some of the HY points that show up on the NBME/USMLE Step 1.

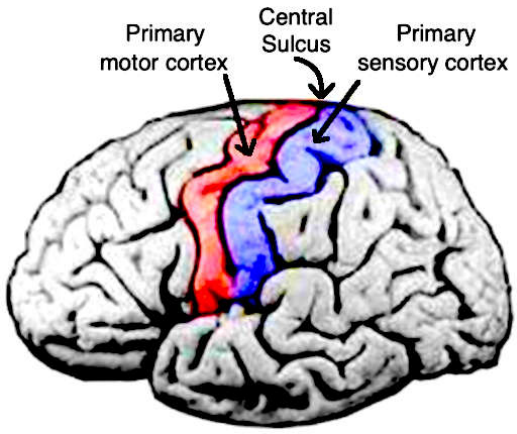
- 27F + long-standing Hx of seizures + the following picture represents her condition; Dx?



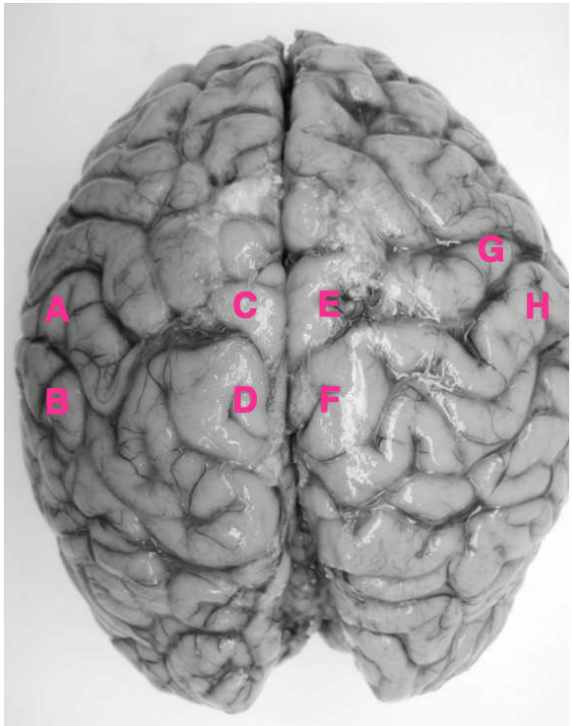
  - Answer = arteriovenous malformation (AVM) → looks almost like swiss cheese. This is a HY image for Step 1.
- 82M + long-standing HTN + weakness of right arm and right side of face; occlusion of which vessel is responsible for patient's presentation? → answer = left middle cerebral artery (MCA) → MCA strokes will cause deficits of contralateral upper limb and face; Hx of HTN suggests carotid atherosclerosis with a plaque launching off to the circle of Willis.
- 74F + atrial fibrillation + weakness of left leg; occlusion of which vessel is responsible for patient's presentation? → answer = right anterior cerebral artery (ACA) → ACA strokes will cause deficits of contralateral lower limb; Hx of AF suggests left atrial mural thrombus launched off to circle of Willis.
- 66M + HTN + weakness of left arm; where is the stroke lesion? (answer the following three Qs)
  - 1) left or right side of the brain? → answer = right (contralateral).
  - 2) medial or lateral cerebral hemisphere? → answer = lateral (homunculus for upper limb and face is lateral; leg is medial).



- 3) anterior or posterior to the central sulcus? → answer = anterior (primary motor cortex is anterior; primary sensory is posterior).

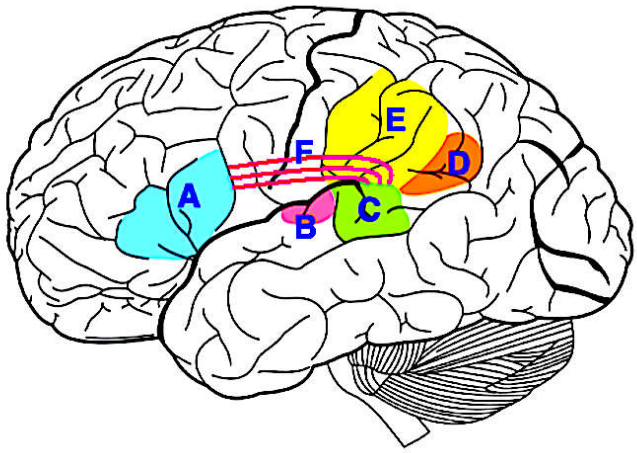


- 92F + atrial fibrillation + numbness/tingling of her right leg; where is the stroke lesion?
  - 1) left or right side of the brain? → answer = left (contralateral).
  - 2) medial or lateral cerebral hemisphere? → answer = medial (homunculus for lower limb is medial; upper limb and face are lateral).
  - 3) anterior or posterior to the central sulcus? → answer = posterior (primary sensory cortex is posterior; primary motor is anterior).
- 66M + chronic HTN managed with lisinopril + paresthesias of left leg; using the following diagram, which letter corresponds to the part of the brain that is affected?



- Answer = F → stroke affects left side of body, is sensory, and is lower limb, so we know that the lesion must be contralateral (i.e., we can eliminate A-D, which are the left side of the brain; i.e., the stroke must be on the right side of the brain), medial (lower limb is medial brain as per the homunculus; upper limb and face are lateral brain), and posterior to the central sulcus (primary sensory cortex is posterior; primary motor cortex is anterior).

- Using the above image, describe how each letter corresponds to a stroke in terms of 1) side of the body affected, 2) sensory vs motor, and 3) upper limb/face vs lower limb:
  - A → Motor disturbance (i.e., paresis/paralysis) of right arm + face.
  - B → Sensory disturbance (i.e., numbness, paresthesias, pain) of right arm + face.
  - C → Motor; right leg.
  - D → Sensory; right leg.
  - E → Motor; left leg.
  - F → Sensory; left leg.
  - G → Motor; left arm + face.
  - H → Sensory; left arm + face.
- 91M + Hx of atrial fibrillation + new-onset difficulty articulating speech (i.e., words come out "telegraphic"); repetition is impaired; comprehension is intact; where on the following diagram is affected? Also describe what lesions of the other letters would result in:



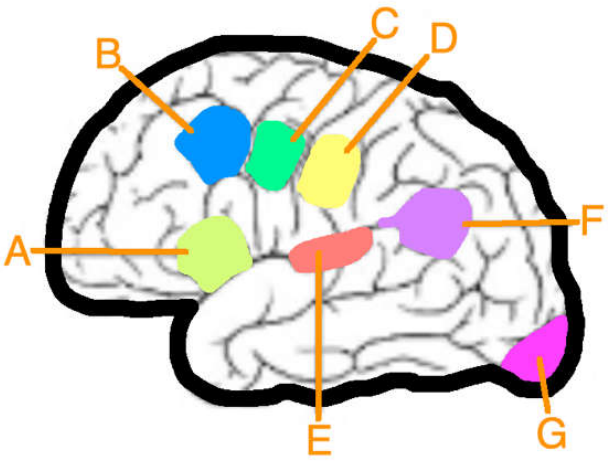
  - Answer = A (Broca area; inferior frontal gyrus; **frontal lobe)** → Broca aphasia → non-fluent aphasia **with repetition impaired**; patient comprehends without a problem and appears frustrated with inability to articulate fluently.

- Choice B (primary auditory cortex) → not HY, but labeled nevertheless because NBME Q might show you this lettered distribution and can be hard to distinguish B from C.
- Choice C (Wernicke area; superior temporal gyrus; **temporal lobe)** → Wernicke aphasia → fluent aphasia **with repetition impaired** → "word salad," where patient can articulate without issue but makes no sense; also cannot comprehend.
- Choice D (angular gyrus; **parietal lobe)** → transfers information to Wernicke area; lesion causes **Gerstmann syndrome:** agraphia, right-left agnosia, finger agnosia, acalculia; not an overwhelmingly HY diagnosis, but students tend to carry an infatuation with it nevertheless.
- Choice E (supramarginal gyrus) → not HY, but labeled nevertheless because NBME Q might show you this lettered distribution; thought to play a role in empathy and interpreting people's gestures.
- Choice F (arcuate fasciculus) → conduction aphasia → impaired repetition only.
- If stroke sounds like Broca but repetition **not** impaired, Dx? → answer = transcortical motor aphasia.
- If stroke sounds like Wernicke but repetition **not** impaired; Dx? → answer = transcortical sensory aphasia.
- If stroke sounds like both Broca + Wernicke at the same time but repetition **not** impaired, Dx? → answer = mixed transcortical aphasia.
- If stroke sounds like both Broca + Wernicke at the same time and repetition **is** impaired; Dx? → answer = global aphasia.

- 72M + Broca aphasia; which vessel is affected? → answer = left MCA (always assume left brain is dominant, unless Q specifically explicates the patient is left-handed).
- 72M + Wernicke aphasia; which vessel is affected? → answer = left MCA (same).
- 72M + Broca aphasia; where in the brain is affected? → answer = inferior frontal gyrus of frontal lobe (reiterating this because of yieldness for Step 1).
- 72M + Wernicke aphasia; where in the brain is affected? → answer = superior temporal gyrus of temporal lobe (reiterating this because of yieldness for Step 1).
- 72M + cannot draw clockface properly; Dx + where is the stroke? → answer = hemineglect syndrome caused by stroke in **non-**dominant parietal lobe (i.e., right MCA in most people) → leads to neglect of

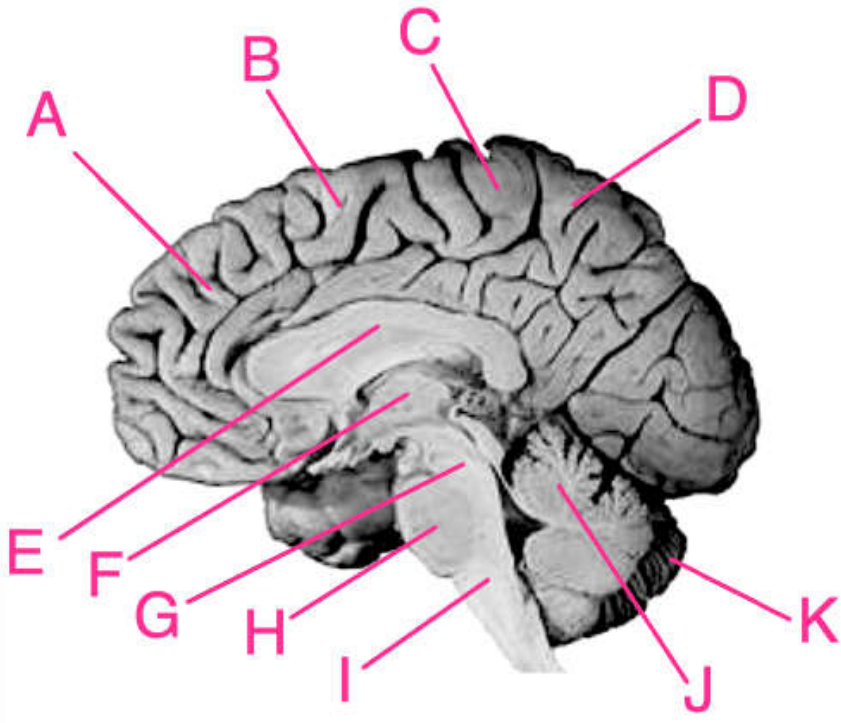
left visual field, although the patient can still see perfectly fine; hemineglect usually affects left visual field (i.e., left clockface) because most brains are left-, not right-, dominant; hemineglect of the right visual field is rare because of redundant processing.

- Identify:



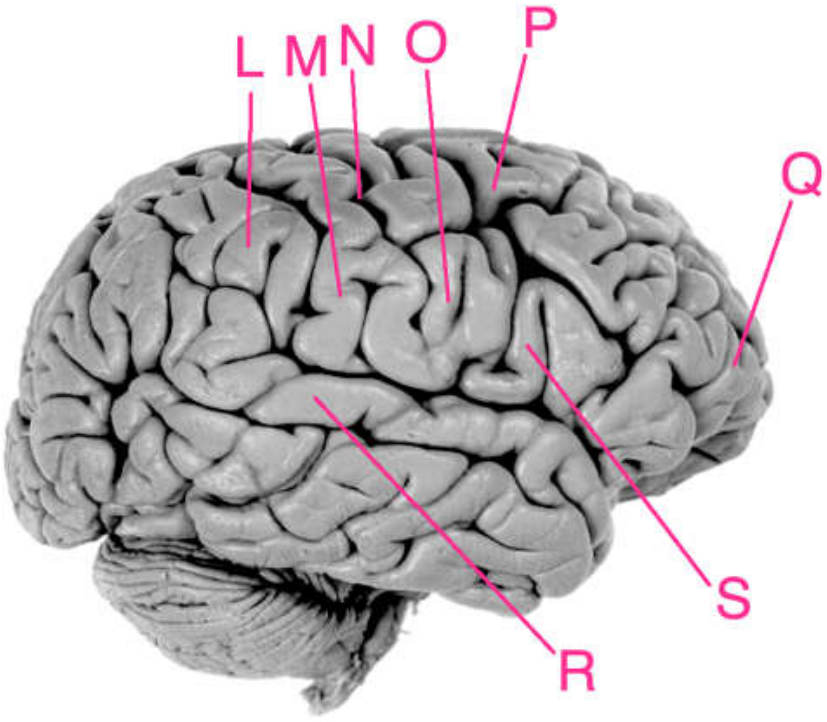
  - A = Broca area; B = premotor cortex; C = primary motor cortex; D = primary sensory cortex; E = primary auditory cortex; F = Wernicke area; G = occipital pole of occipital lobe.

- Identify:



- A = frontal lobe;
- B = medial premotor cortex; the answer if the vignette mentions the patient **planning on** making a movement of his left leg.
- C = medial primary motor cortex (pre-central gyrus); the answer if post-stroke patient has weakness of the left leg.
- D = medial primary sensory cortex (post-central gyrus); the answer if the post-stroke patient has sensory disturbance (i.e., paresthesias, numbness) of the left leg.
- E = corpus collosum; F = thalamus; G = midbrain; H = pons; I = medulla.
- J = cerebellar vermis; the answer if patient has truncal ataxia.
- K = posterior lobe of cerebellum; the answer if the patient has right limb ataxia **(ipsilateral for cerebellar lesions).**

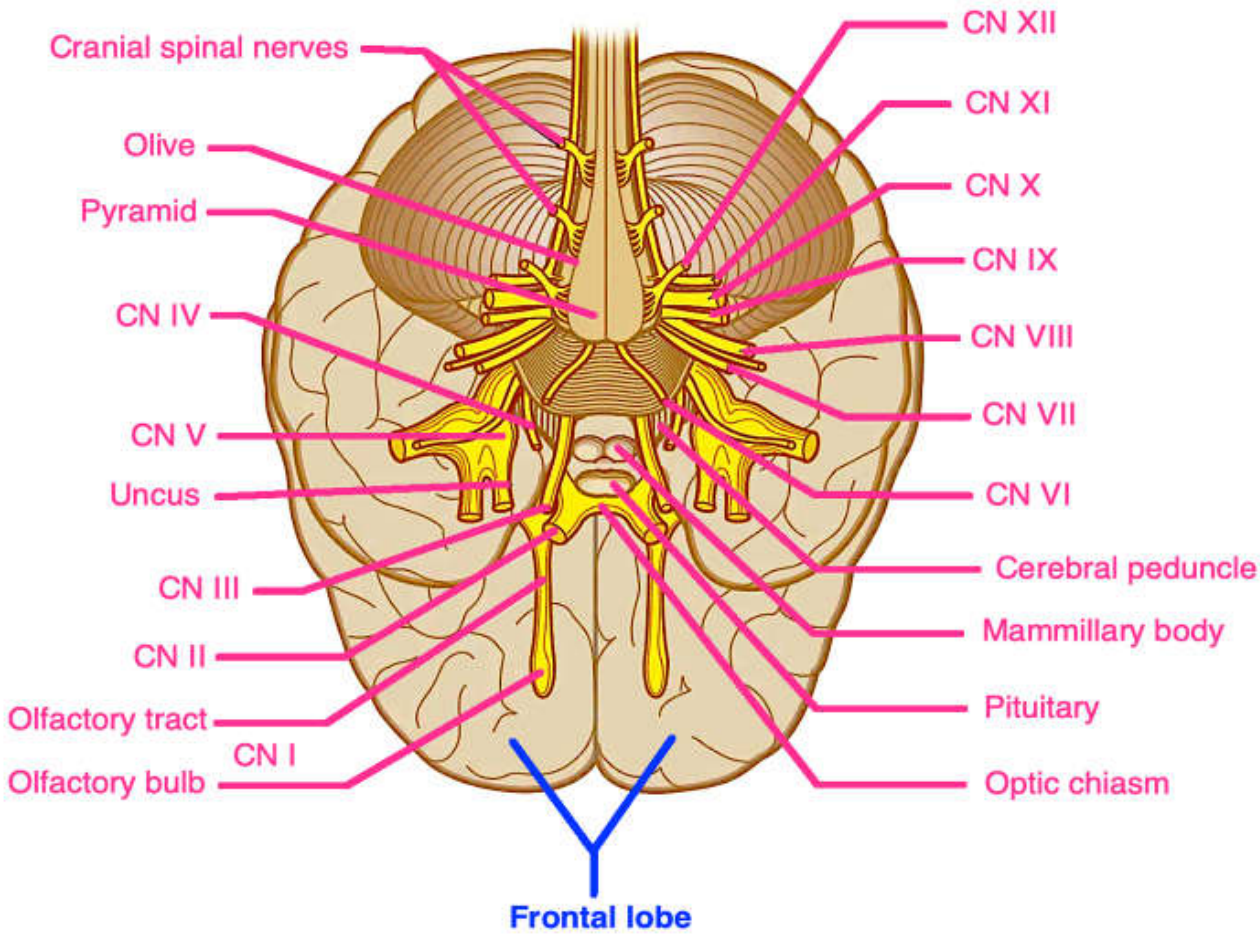
- Identify:



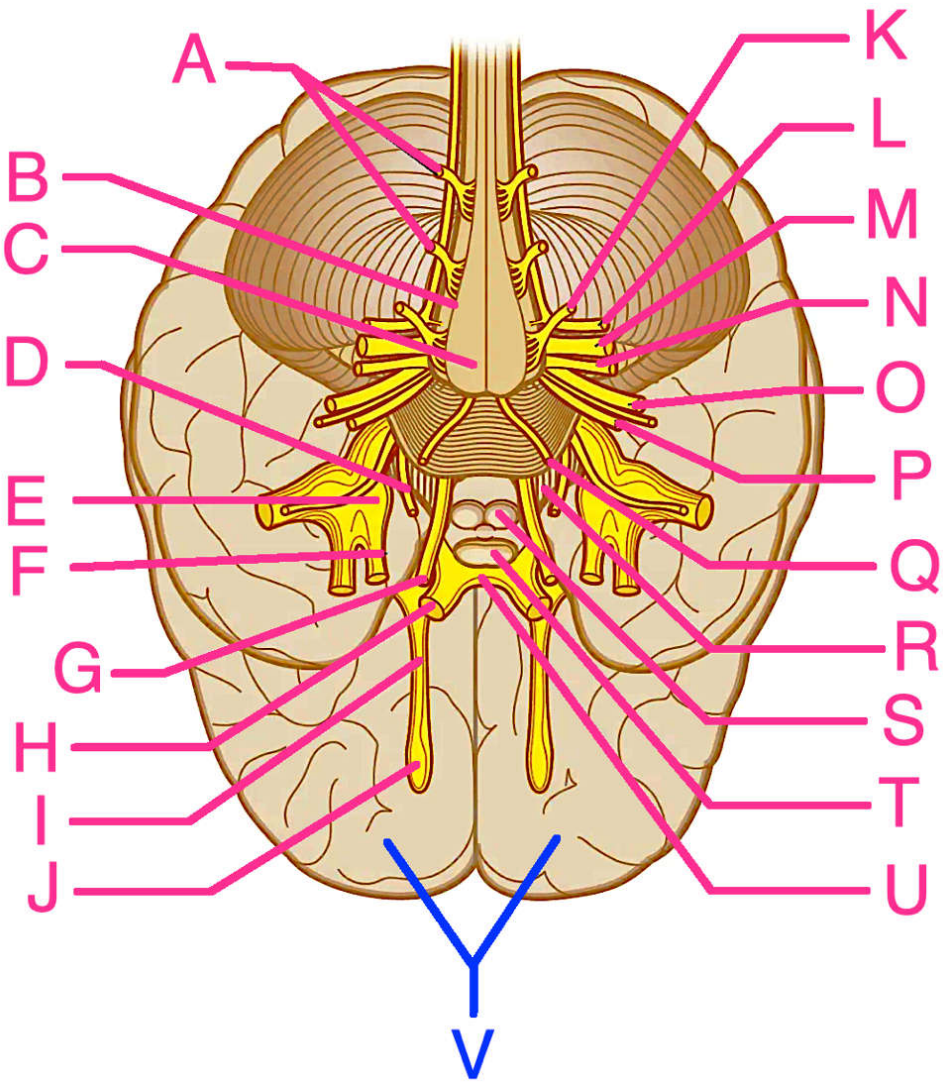
- L = somatosensory association area; integrates information received from the primary sensory cortex (lies just posterior to primary sensory cortex).
- M = lateral primary sensory cortex (post-central gyrus); the answer if the post-stroke patient has sensory disturbance (i.e., paresthesias, numbness) of the left arm and face.
- N = central sulcus.

- O = lateral primary motor cortex (pre-central gyrus); the answer if post-stroke patient has weakness of the left arm and face.
- P = medial premotor cortex; the answer when planning on moving the left leg.
- Q = frontal lobe; the answer for post-stroke patient with apathy, disinhibition, and personality change.
- R = superior temporal gyrus of non-dominant hemisphere (in most people, the left superior temporal gyrus is dominant and contains the primary auditory cortex and Wernicke area).
- S = lateral premotor cortex; the answer when planning on moving the left arm + face.

- First learn:

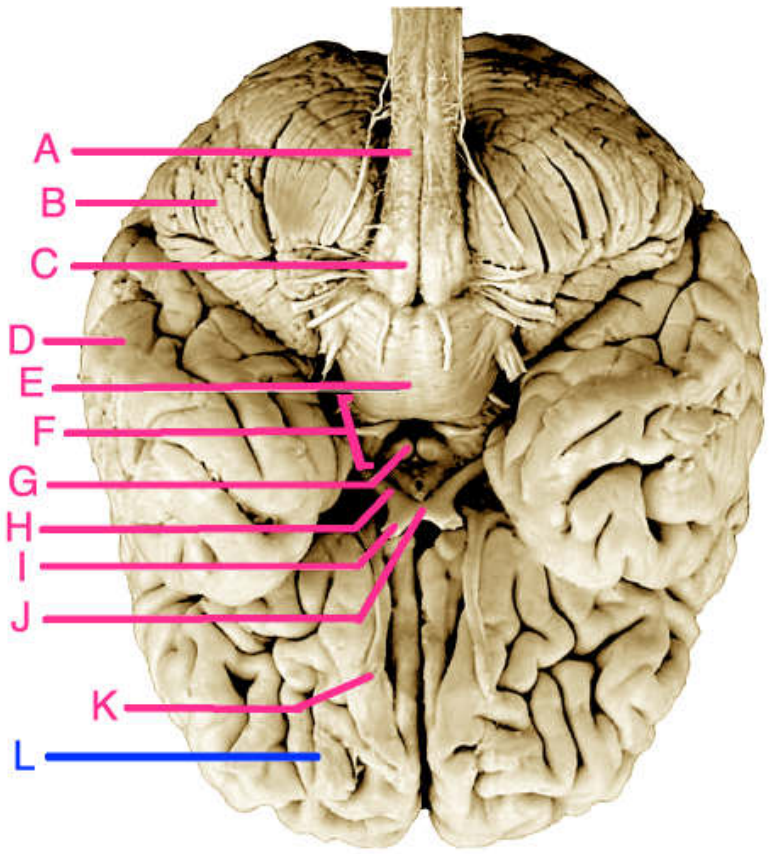


- Now identify:

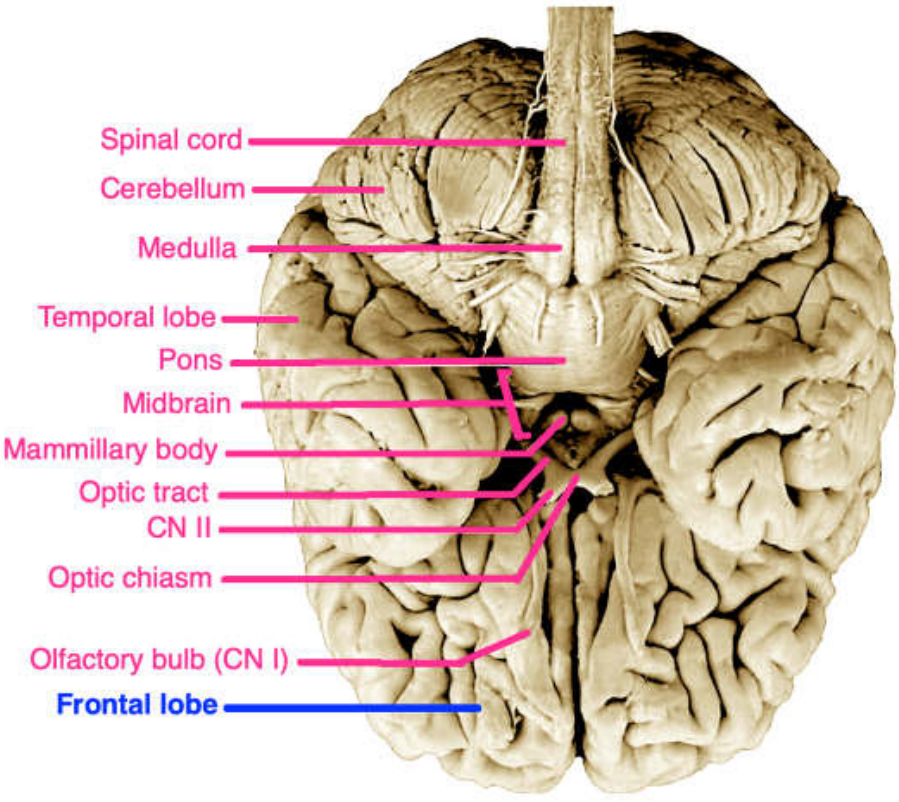


- Frontal lobe shown in blue because of its high yieldness. In other words, **the USMLE will show you a gross specimen without the cerebellum,** where it can be difficult to tell which side is the frontal lobe. Therefore, recognizing the olfactory bulbs/tracts and saying "Ok, that's the frontal lobe" is really important.

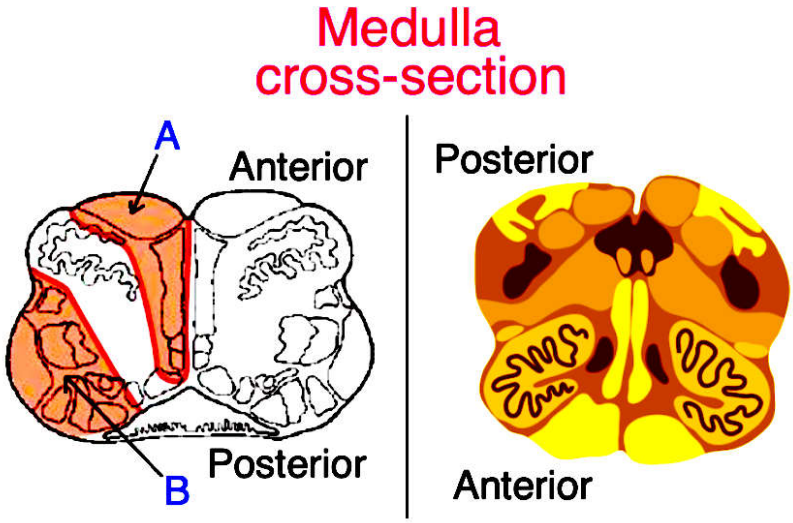
- 72M + progressive personality change + apathy + pulls pants down when guests come over to the house; where on the following specimen is most likely the location of the pathology in this patient?



- Answer = L (frontal lobe); Dx = frontotemporal dementia (classically apathy, disinhibition, personality change).

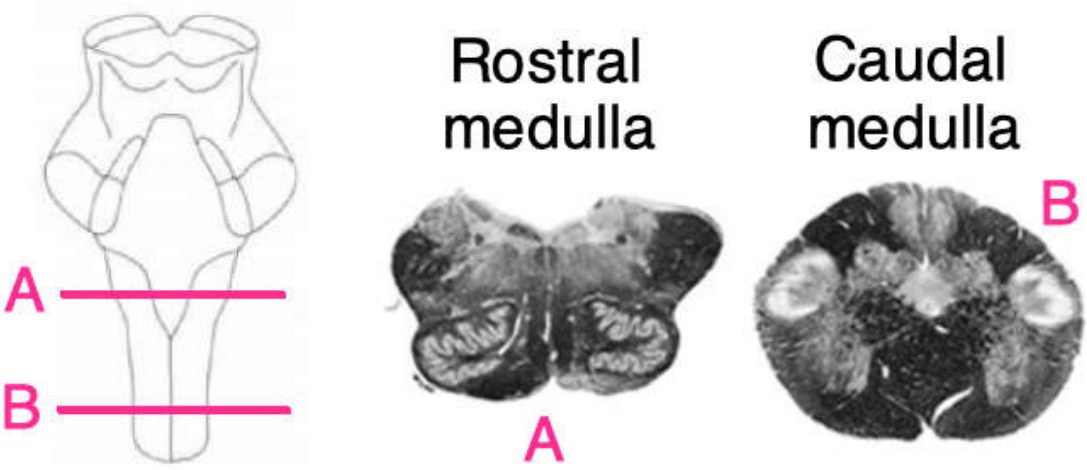


- 69F + Hx of HTN + stroke + tongue deviated to left side; which of the following letters corresponds on the stroke lesion?



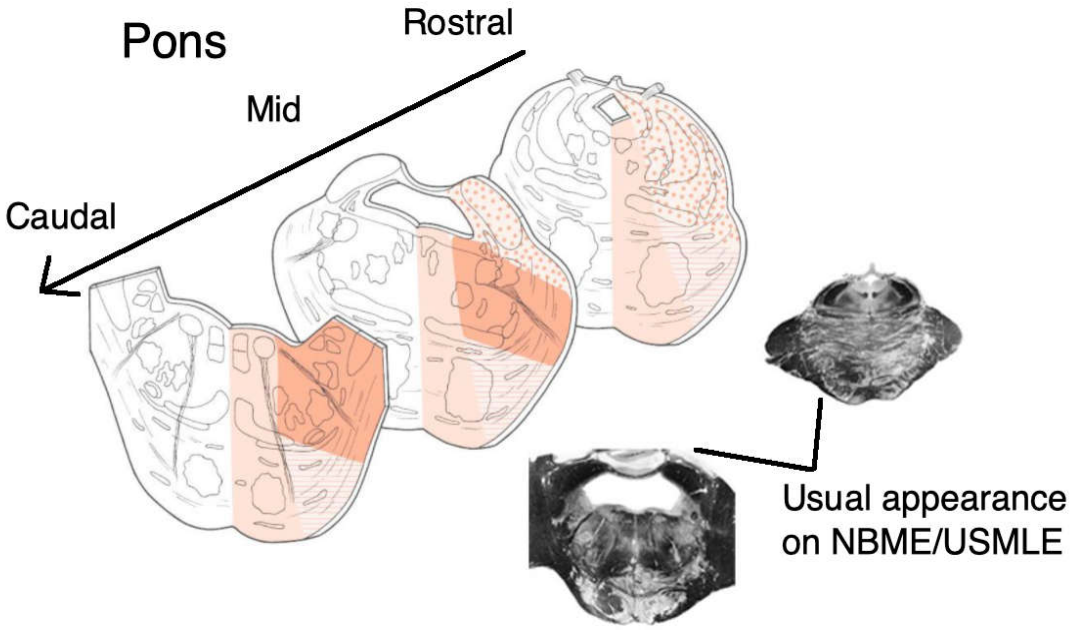
  - Most students: "wtf am I looking at?" Yeah I get, but stick with me here. Answer = A → medial medullary syndrome. The NBMEs for Step 1 are littered with these types of cross-sections. You **do not** need to know every structure labeled on a cross-section. Waste of your time. You merely need to know that **ipsilateral tongue deviation = medial medullary syndrome.** So if tongue deviation is to the left, you say, "Ok, well that must be medial medulla of the same side, so I'll just choose the letter on the left side of the cross-section that is most medial." This is supplied by the **paramedian branch of the anterior spinal artery (ASA).**
    - In short: if tongue deviation in vignette; answer = medial medulla of ipsilateral side.
  - Choice B refers to lateral medullary syndrome (Wallenberg syndrome) → patient will have three main features: **hoarseness of voice + dysphagia +/- Horner syndrome.** You might say, "Wait, I thought Horner syndrome was for Pancoast tumor of the lung." Yeah, but it's also sometimes seen in lateral medullary syndrome. This is a huge giveaway if you get it in the vignette. You can also remember "PICACHEW" (Pikachu from Pokemon) → **PICA → posterior inferior cerebellar artery is affected** + CHEW → dysphagia. **If PICA not listed, choose vertebral artery.**
    - In short: if hoarseness/dysphagia +/- Horner in vignette, answer = lateral medulla of ipsilateral side.

- How the medulla cross-sections tend to appear on the Step 1 NBME/USMLE:

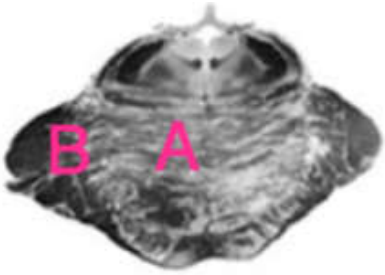


  - If you get these types of slices on your exam and are deer in the headlights (i.e., you have no idea what you're looking at), just remember: pay attention to whether they give you tongue deviation versus hoarseness/dysphagia +/- Horner; for tongue deviation, just choose the ipsilateral medial part of whichever slice they give you; for hoarseness/dysphagia +/- Horner, just choose ipsilateral lateral part of whichever slice they give you. No need to memorize 50 random structures.

- How does the pons tend to appear on the Step 1 NBME/USMLE?

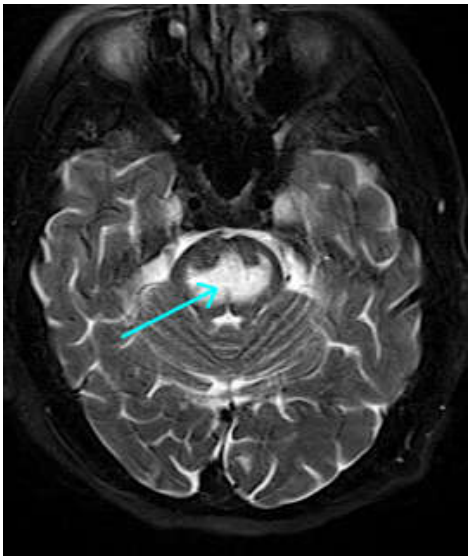


- 70M + Hx of HTN + left-sided Bell's palsy + left-sided hearing loss + loss of temperature/pain to left side of face and right side of body; which of the following letters corresponds to the stroke lesion?



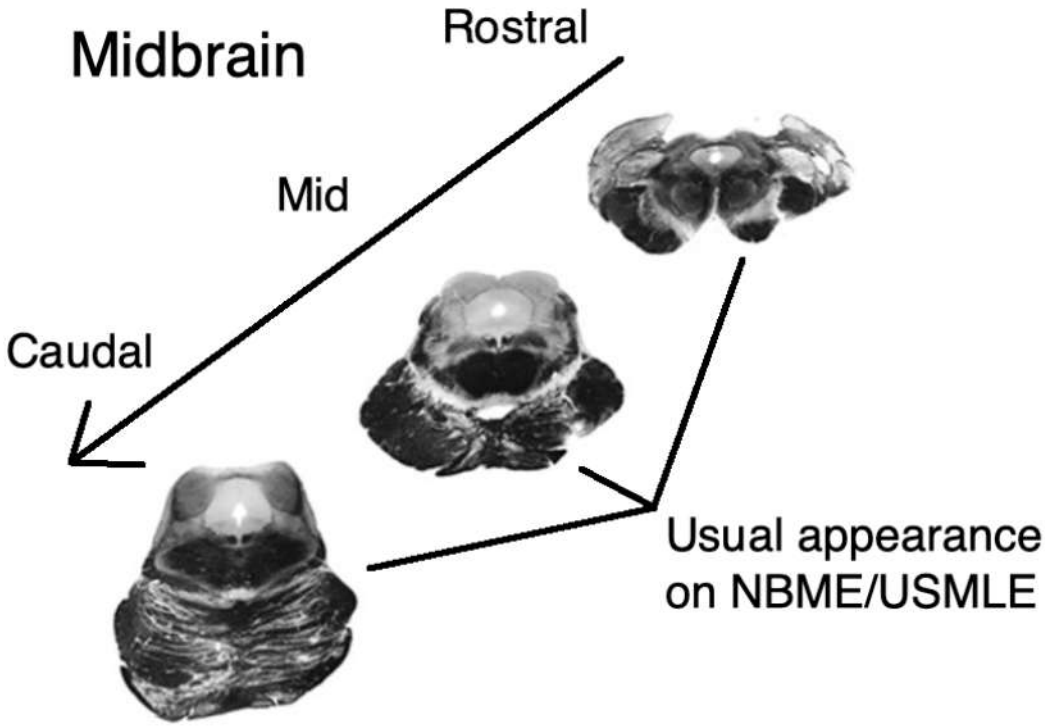
- Answer = B → lateral pontine syndrome → **classically causes ipsilateral Bell's palsy** +/- hearing loss; the Bells palsy is literally the key detail you need to know. Caused by lesion of **anterior inferior cerebellar artery (AICA)** → mnemonic is FACIAL (Face is affected, and the word FACIAL contains AICA backwards).
- Choice A = medial pontine syndrome; for some reason lower yield compared to the aforementioned brainstem lesions, but be aware that it can cause abducens nerve palsy, so patient can have esotropia (medially directed eye).
- Bottom line is: if you get a question where it's a stroke of some kind and the patient has Bell's palsy, just choose the lateral letter on the ipsilateral side of the pons slice.

- Looking at the following CT of the pons, what's the diagnosis; and what was the most likely cause?

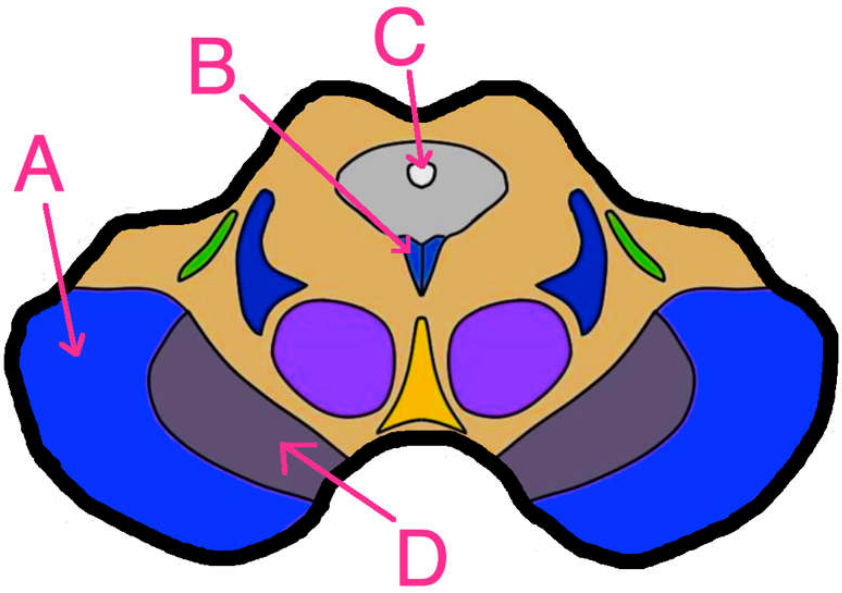


- Diagnosis is central pontine myelinolysis secondary to correction of hyponatremia too rapidly with hypertonic saline; the patient will present with locked-in syndrome (inability to move entire body except for eye muscles; described as triad of quadriplegia + aphonia + impaired horizontal eye movements); do not confuse this with cerebral edema caused by correcting hypernatremia too quickly with hypotonic saline.

- How does the midbrain tend to appear on the Step 1 NBME/USMLE?



- Identify:

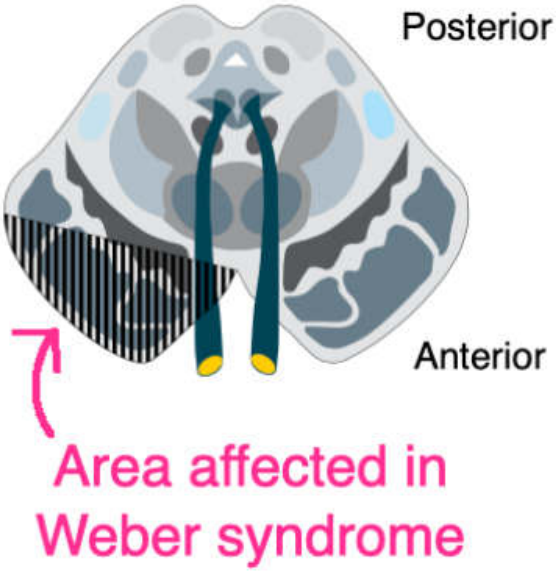


  - A = crus cerebri; B = CN III (oculomotor) nucleus; C = cerebral aqueduct; D = substantia nigra.
  - HY point: crus cerebri lesions cause contralateral spastic paralysis.

- 78M + Hx of HTN + left eye is directed laterally and inferiorly + weakness of right side of the body; Dx?

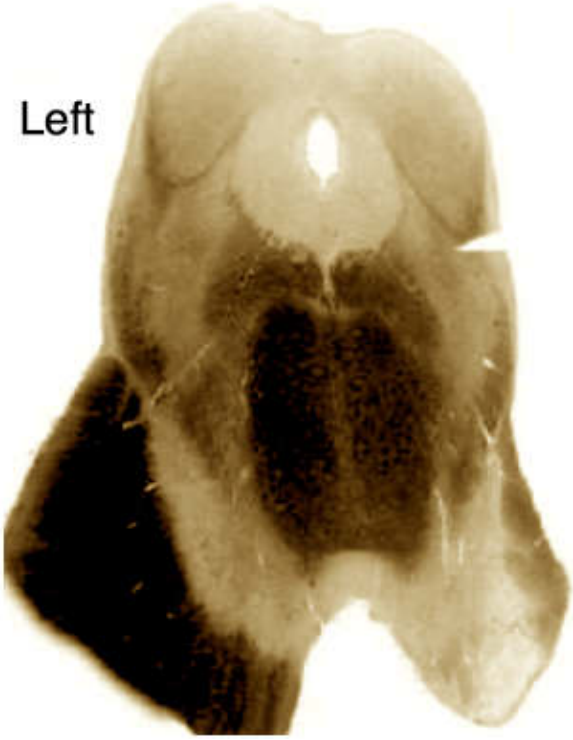
  → NBME answer = Weber syndrome → **midbrain** lesion resulting in ipsilateral oculomotor nerve

(CNIII) palsy + contralateral hemiparesis (loss of crus cerebri); lesion in paramedian branch of posterior cerebral artery (PCA).



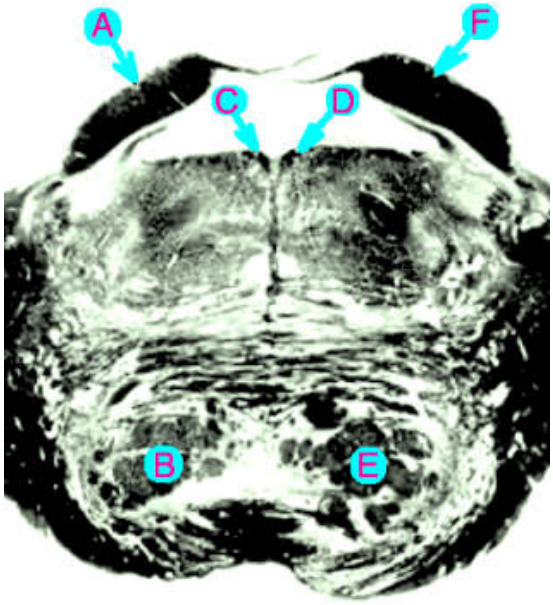
- Bottom line is: if the vignette gives you a stroke syndrome of some kind where there's oculomotor nerve palsy, choose Weber syndrome (midbrain lesion).

- 75F + many years ago had cerebral infarction + brain on autopsy appears as follows; what would her main deficit have been?



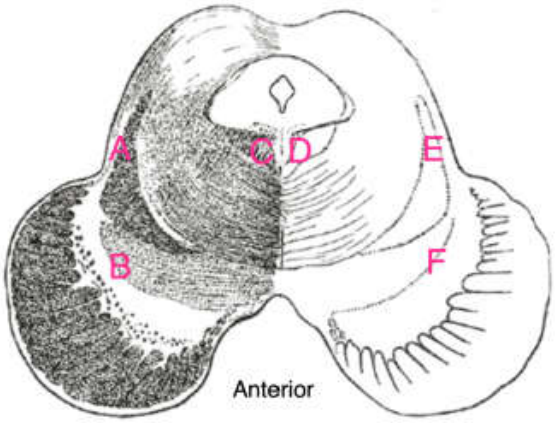
- Answer = spastic paralysis on the left side of the body (right crus cerebri infarcted).

- Identify:

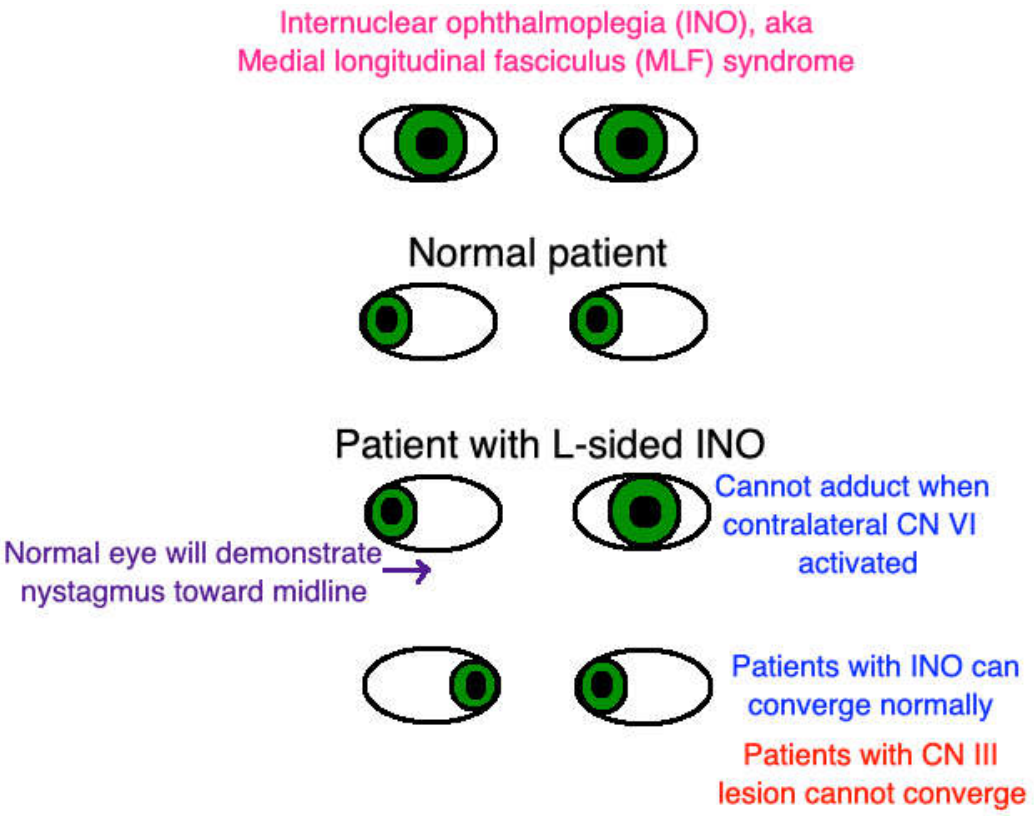


- A = right inferior colliculus; B = right pyramidal tract; C = right medial longitudinal fasciculus (MLF); D = left MLF; E = left pyramidal tract; F = left inferior colliculus.
- The inferior colliculi are part of the auditory system and play a role in sound processing.
- The MLF are the answer in multiple sclerosis when the patient cannot adduct one eye while abducting the other – for instance, choice D would be the answer if the patient cannot adduct the left eye while abducting the right; the right eye would exhibit nystagmus.

- 32F + one-week Hx of urge to run to the bathroom at inopportune moments + physical exam shows inability to adduct left eye when gazing to the right + convergence of gaze not affected; where on the following diagram represents the location of the pathology in this patient?



- Answer = D; diagnosis is medial longitudinal fasciculus syndrome (MLF syndrome; aka internuclear ophthalmoplegia; INO) in a patient with multiple sclerosis (vignette also describes urge incontinence); normal convergence tells us that CN III is intact bilaterally.
- What you need to know is this: in INO, the side that cannot adduct is the side of the pathology, so we know all letters on the patient's right (A-C) are wrong; so looking at D-F, even if you have no idea, if you recognize the presentation as MLF syndrome, you can say, "Well, the name of the condition is **medial** longitudinal fasciculus syndrome, so I'll just choose the letter that's **medial."**



- Mechanism for pathology: when you look to one side (let's say the right), CN VI on the right and CN III on the left are activated. This unison of activation is accomplished when the left MLF is activated. In the case of MLF syndrome, the right eye can abduct without a problem, but the left eye will fail to adduct and the right eye will exhibit nystagmus back toward the midline. **MLF syndrome is pathognomonic for multiple sclerosis.** MS patients can also get optic neuritis (i.e., change in visual acuity, color vision, central scotoma, etc.), but this is not

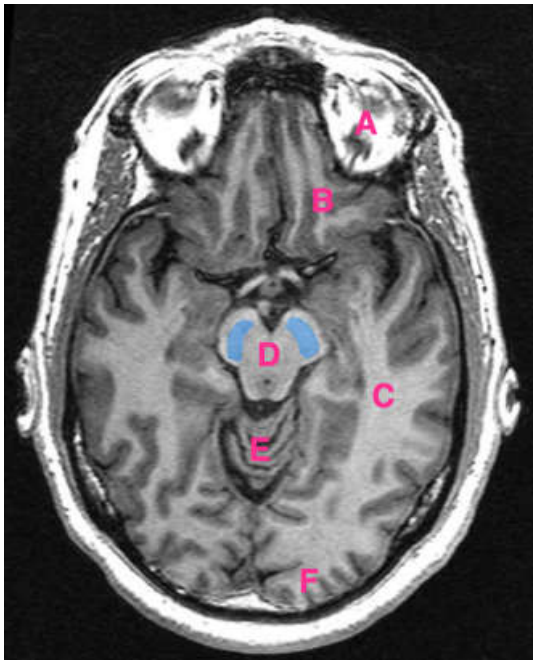
pathognomonic for MS (i.e., it can be seen in other circumstances, such as with ethambutol or sildenafil use).

- Exceedingly rare but MLF syndrome may occur in stroke due to HTN (still considered pathognomonic for MS).

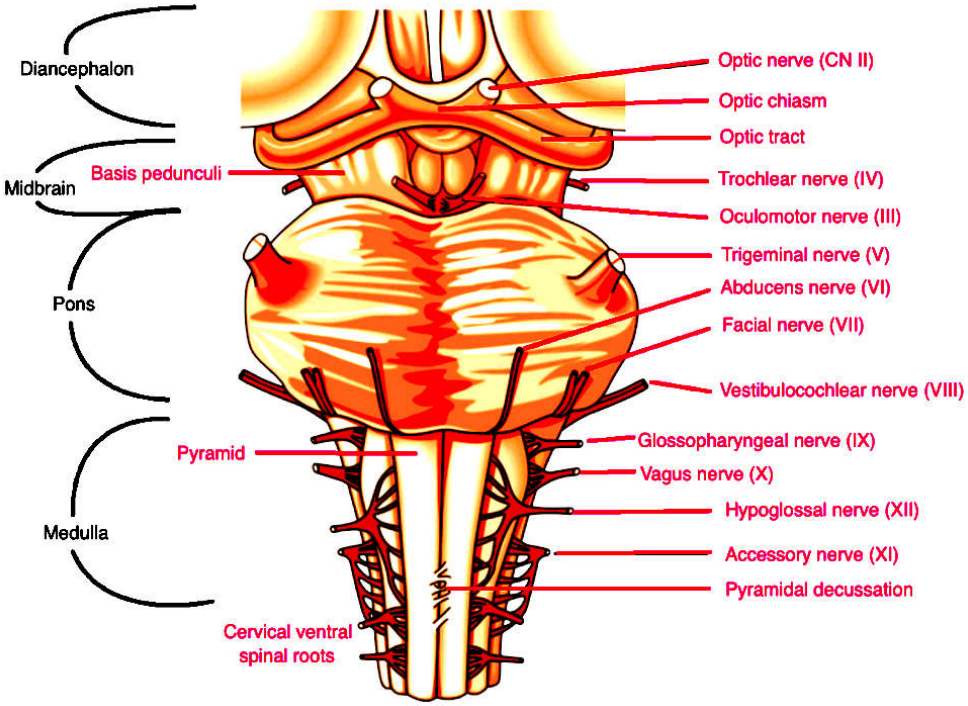
- 74M + unilateral resting tremor + shuffling gait; where on the following MRI be represents the location of his pathology?



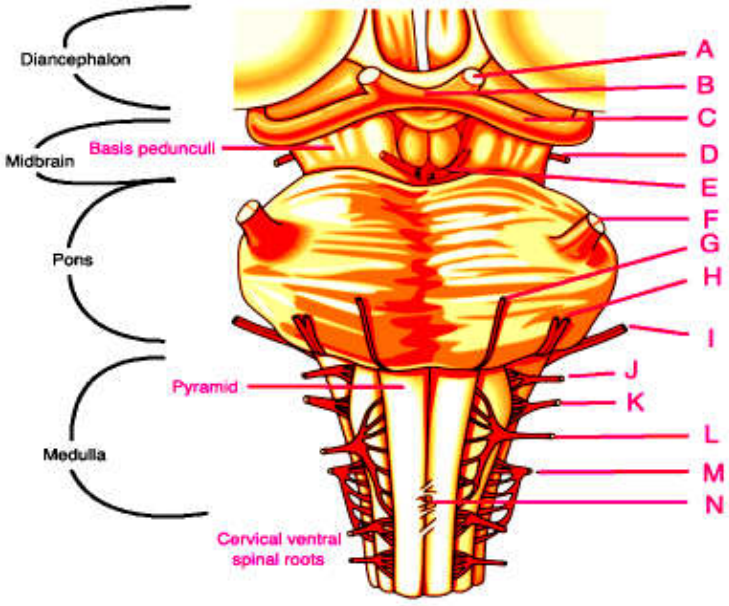
- Answer = D (midbrain); the substantia nigra pars compacta (blue) are located bilaterally within the midbrain:



- Choice A = optic nerve; B= medial orbital gyrus; C = parietal Lobe; D = midbrain (substantia nigra pars compacta in blue); E = cerebellar vermis; F = occipital pole (of occipital lobe).
- "Do I need to know the cranial nerves coming off the brainstem?" → Yeah. NBME/USMLE will show you a brainstem specimen or drawing, and you need to be able to identify the nerve(s) they're referring to:

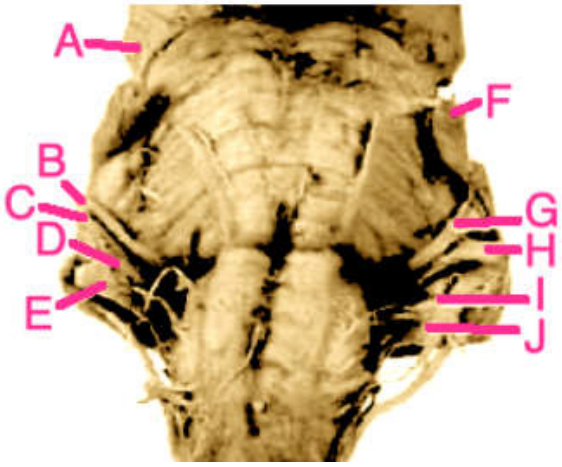


- 32F + hiking in Connecticut + circular rash on forearm + Bell's palsy; which nerve on the following diagram explains her findings?



- Answer = facial nerve (choice H) → Bell's palsy secondary to Lyme disease.

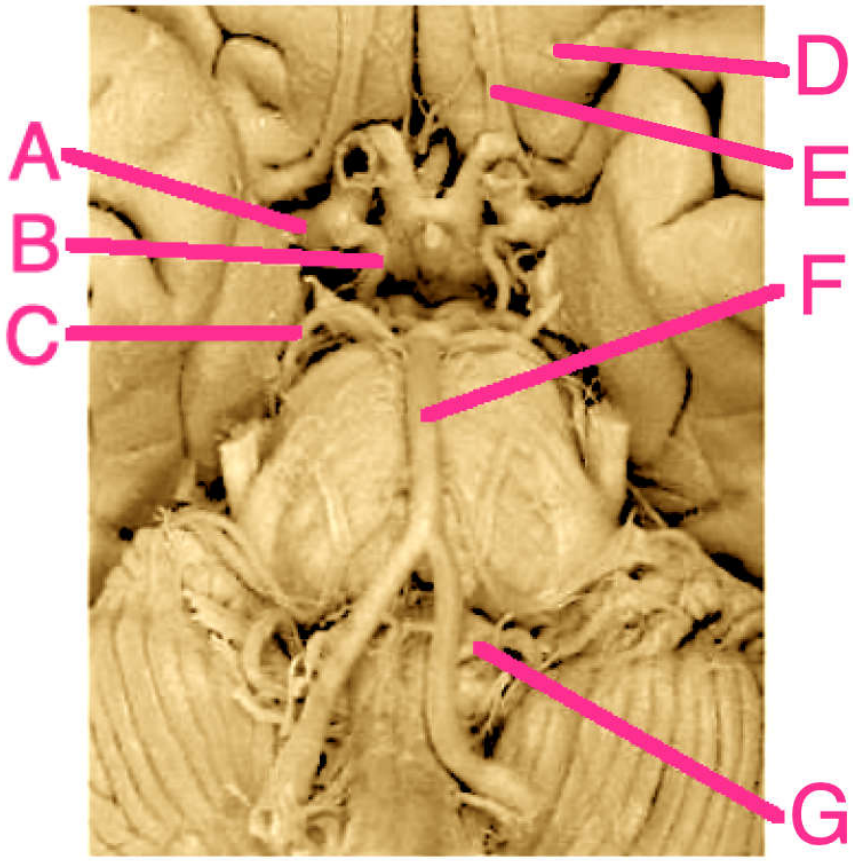
- 76M + stroke + uvula deviates to the left when he says "Ahh"; which letter on the following image is the most likely location of this patient's pathology?



- Answer = right vagus nerve (choice E); uvula deviates to contralateral side; Choice A = right trochlear nerve; B = right facial nerve; C = right vestibulocochlear nerve; D = right glossopharyngeal nerve; F = left trigeminal nerve; G = left facial nerve; H = left vestibulocochlear nerve; I = left glossopharyngeal nerve; J = left vagus nerve.

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CRANIAL NERVE EXIT POINTS		
Cranial nerve	Exit point	Extra points
Olfactory (CN I)	Cribriform plate	Only CN that doesn't relay through thalamus
Optic (CN II)	Optic canal	**Afferent component of light reflex**
Oculomotor (CN III)	Superior orbital fissure	**Efferent component of light reflex (constricts pupil via parasympathetic innervation)**
Trochlear (CN IV)	Superior orbital fissure	Innervates superior oblique (eye looks inferomedially)
Trigeminal ophthalmic branch (V1)	**S**uperior orbital fissure	Afferent component of corneal reflex
Trigeminal maxillary branch (V2)	Foramen **R**otundum	Trigeminal mnemonic: "**S**tanding **R**oom **O**nly"
Trigeminal mandibular branch (V3)	Foramen **O**vale	Muscles of mastication
Abducens (VI)	Superior orbital fissure	Abducts eye (patient gazes laterally)
Facial (VII)	Auditory canal	Efferent component of corneal reflex; UMN results in contralateral mid + lower facial paralysis (forehead spared); LMN results in ipsilateral Bells palsy
Vestibulocochlear (VIII)	Auditory canal	Can give rise to acoustic schwannoma
Glossopharyngeal (IX)	Jugular foramen	Afferent component of gag reflex + carotid sinus baroreceptors
Vagus (X)	Jugular foramen	Efferent component of gag reflex + parasympathetic to slow HR; lesion causes **contralateral** uvula deviation
Accessory (XI)	Jugular foramen	Innervates sternocleidomastoid + trapezius
Hypoglossal (XII)	Hypoglossal canal	Lesion causes **ipsilateral** tongue deviation

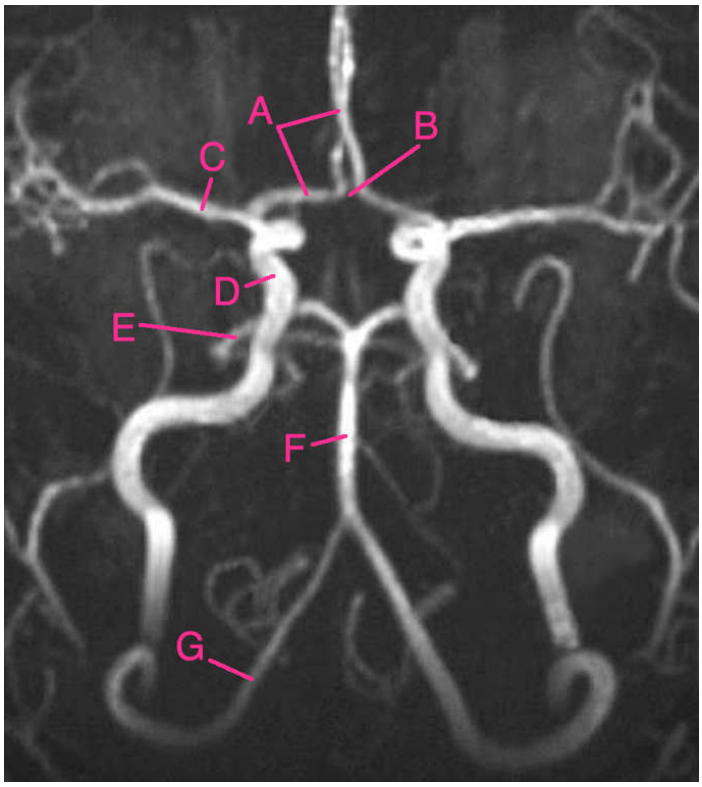
- 35F + complete loss of hearing in left ear; which CN is damaged? → answer = left vestibulocochlear.
- Which nerves enable the gag reflex? → answer = CN IX (afferent); CN X (efferent).
- 47M + touching the face just lateral to the nose produces pain; the nerve supplying this area of the face exits the skull through where? → answer = foramen rotundum (maxillary branch of trigeminal nerve).
- 19M + viral infection + Bell palsy; Q asks what other finding might be seen in this patient as a result of the nerve lesion → answer = right sided hyperacusis (loud sounds) → CN VII (facial) innervates stapedius muscle, which controls the stapes bone and functions to dampen sounds; lesion of CN VII can lead to hyperacusis in some patients.
- 48F + Hx of cystic kidneys + right pupil enlarged; an aneurysm of which of the following vessels might explain this presentation?



  - Answer = posterior communicating artery (choice B); PCoM aneurysms can cause ipsilateral mydriasis (blown pupil); ↑ risk of ACoM and PCoM aneurysms in ADPKD; A = right internal

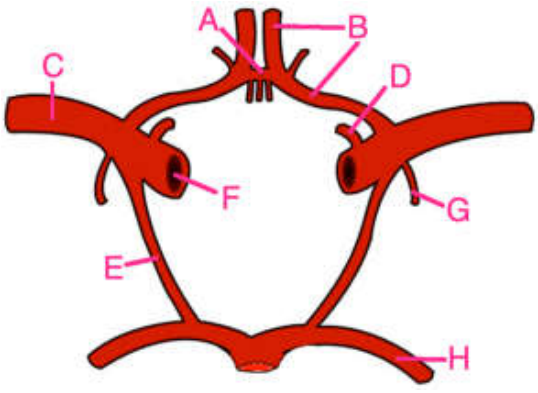
carotid artery; C = posterior cerebral artery (PCA); D = frontal lobe; E = olfactory tract; F = basilar artery; G = posterior inferior cerebellar artery.

- 82F + Hx of atrial fibrillation + inability to move all muscles of body except for eyes; which of the letters on the following angiogram best reflects the location of the patient's stroke?



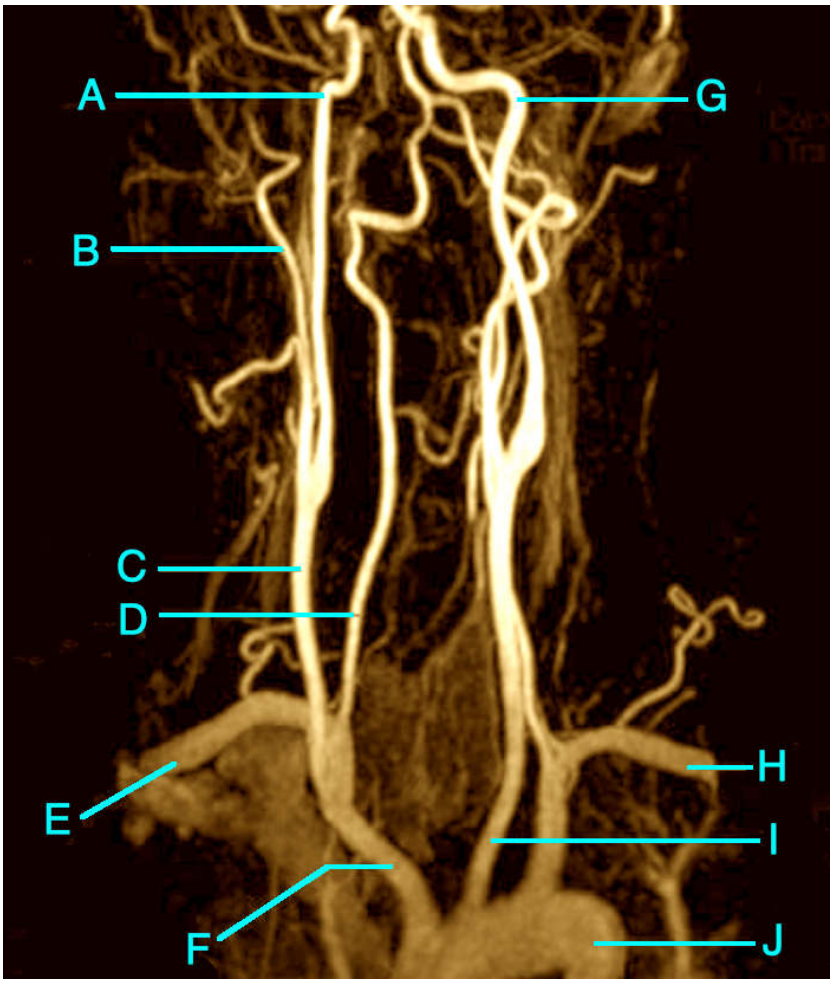
  - Angiogram is of the circle of Willis; answer = basilar artery stroke (choice F), resulting in locked-in syndrome; A = anterior cerebral artery (ACA); B = anterior communicating artery (ACoM); C = middle cerebral artery (MCA); D = internal carotid artery; E = posterior cerebral artery (PCA); G = vertebral artery.

- 68M + Hx of HTN + acute-onset painless loss of vision in the right eye; which letter on the following circle of Willis illustration best reflects the path of the patient's embolus?



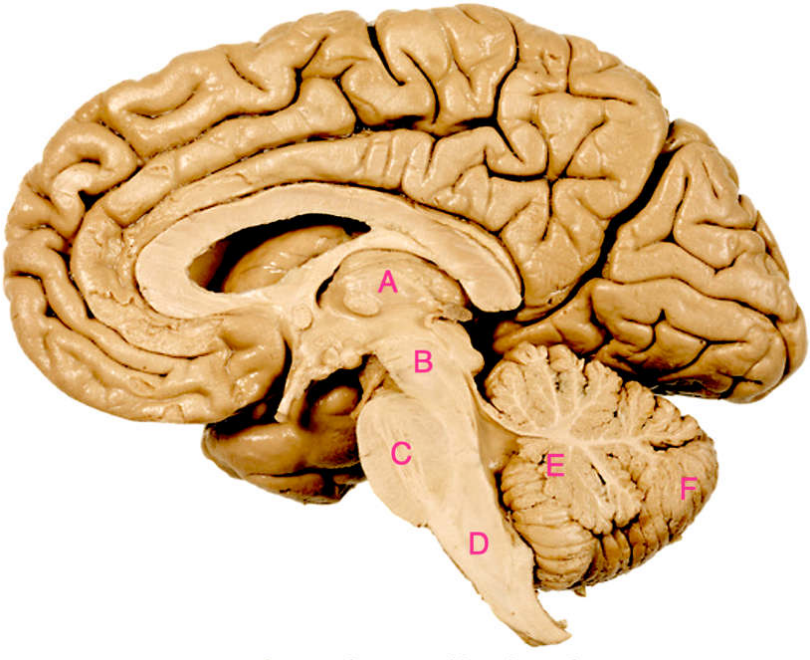
- Answer = D (ophthalmic artery); patient has central retinal artery occlusion from carotid plaque that launched off secondary to HTN; the retinal artery (aka central retinal artery) is a branch of the ophthalmic artery; A = anterior communicating artery (ACoM); B = anterior cerebral artery (ACA); C = middle cerebral artery (MCA); E = posterior communicating artery (PCoM); F = internal carotid artery; G = choroidal artery; H = posterior cerebral artery (PCA).

- 32M + Hx of fainting episodes + blood pressure lower in right arm than left arm; retrograde flow through which vessel in the following angiogram might be expected in this patient?



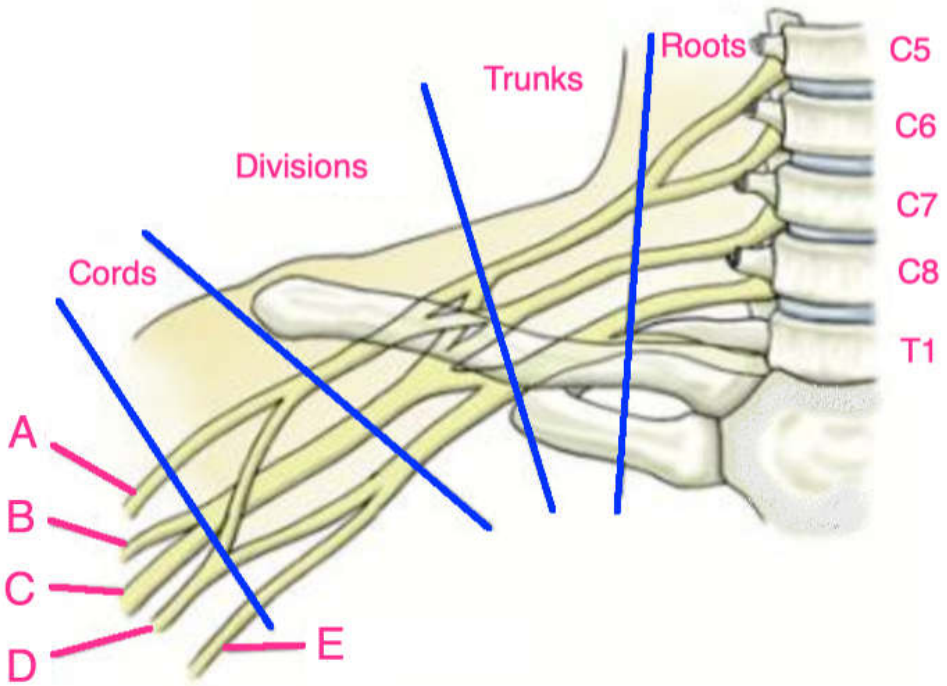
- Answer = right vertebral artery (choice D); diagnosis is subclavian steal syndrome (proximal stenosis of subclavian artery results in retrograde flow through the vertebral artery, which is the first branch of the subclavian). Choice A = right internal carotid artery; B = right external carotid artery; C = right common carotid artery; E = right subclavian artery; F = brachiocephalic artery; G = left internal carotid artery; H = left subclavian artery; I = left common carotid artery; J = aortic arch.

- 34F + boating accident + dysmetria of right side of body + normal strength; which letter on the following sagittal brain cross-section most likely corresponds to the site of her injury?



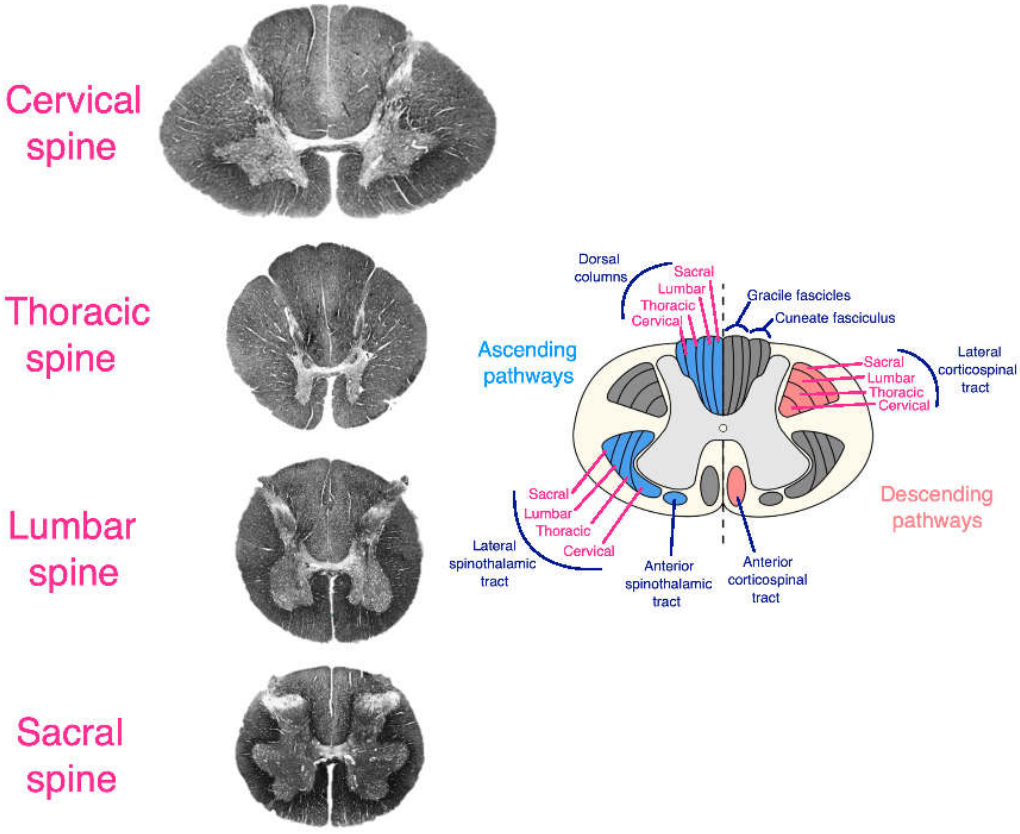
  - Answer = cerebellar hemisphere (choice F; posterior lobe is labeled); cerebellar injuries cause ipsilateral dysmetria; choice E, in contrast, which is the cerebellar vermis, would cause truncal ataxia. Choice A = thalamus; B = midbrain; C = pons; D = medulla.

- 26M + has arm slashed in bar fight + presents with arm pronated and unable to extend wrist; which letter on the following image corresponds to this patient's deficit?

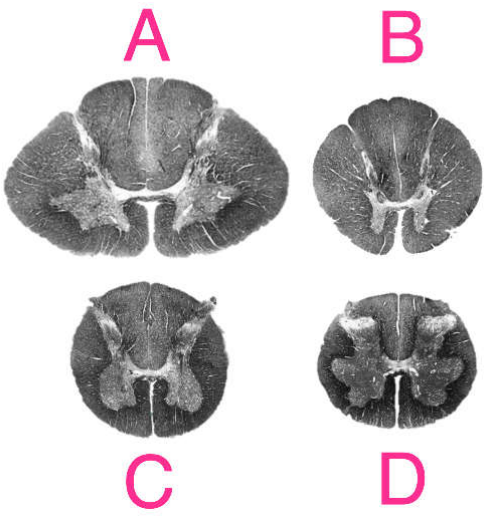


- Answer = radial nerve (choice C). A = musculocutaneous nerve; B = axillary nerve; D = median nerve; E = ulnar nerve.
- USMLE will ask you relatively easy presentation for a neurologic deficit but require you to identify where on the brachial plexus is affected.

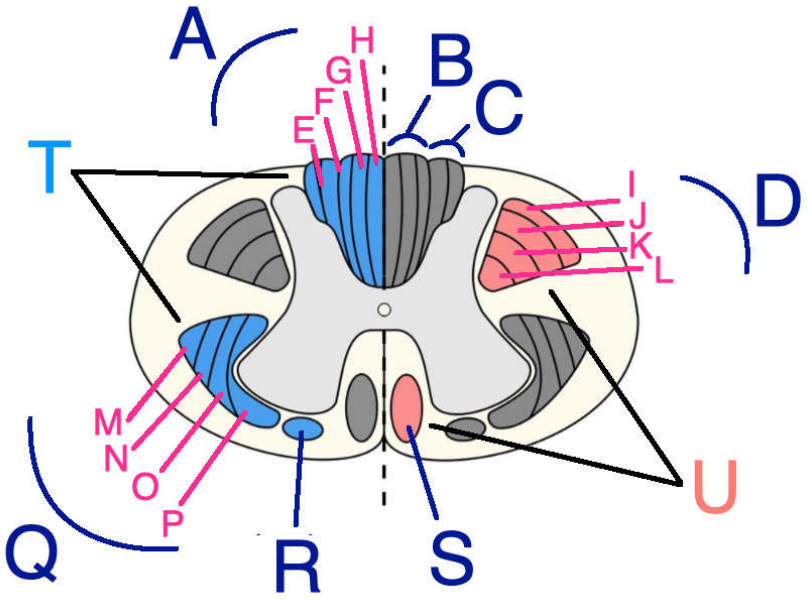
- "What about spinal cord cross-sections?" → Know how following tracts course through the spinal cord:



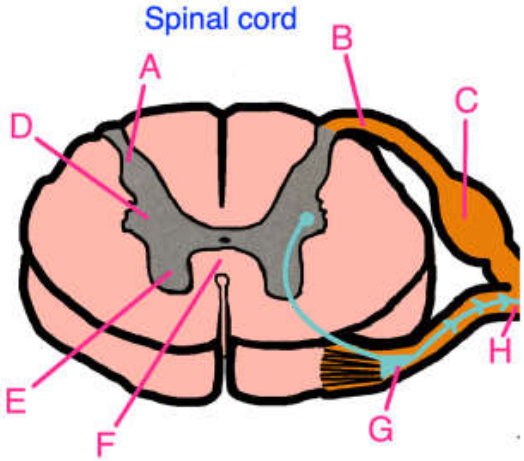
- "Okay, but what do those tracts actually do?"
  - **Spinothalamic → pain and temperature. (HIGH YIELD)**
  - **Dorsal columns → vibration + proprioception. (HIGH YIELD)**
  - **Anterior + lateral corticospinal tracts → motor function. (HIGH YIELD)**
- Identify:



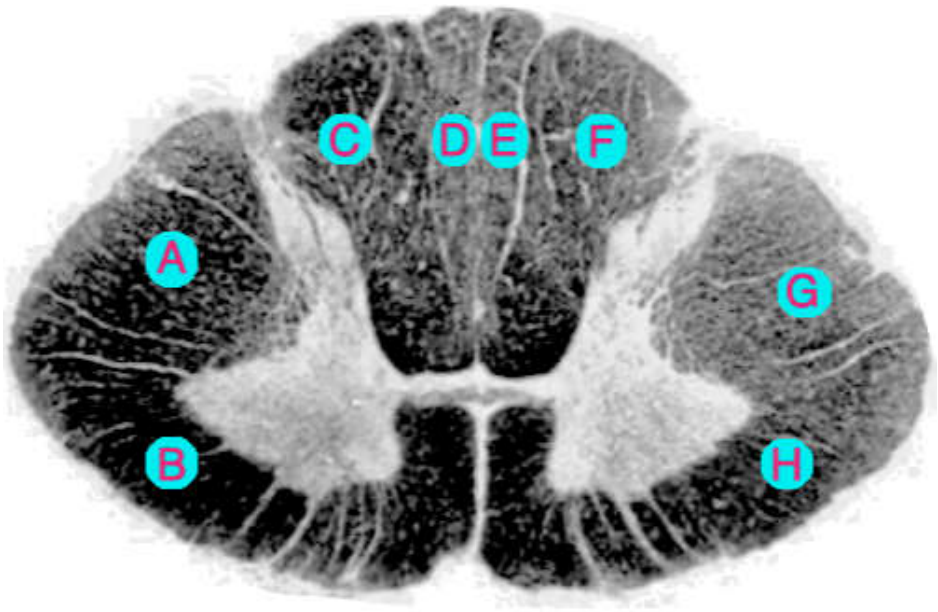
- Identify:



- Identify:

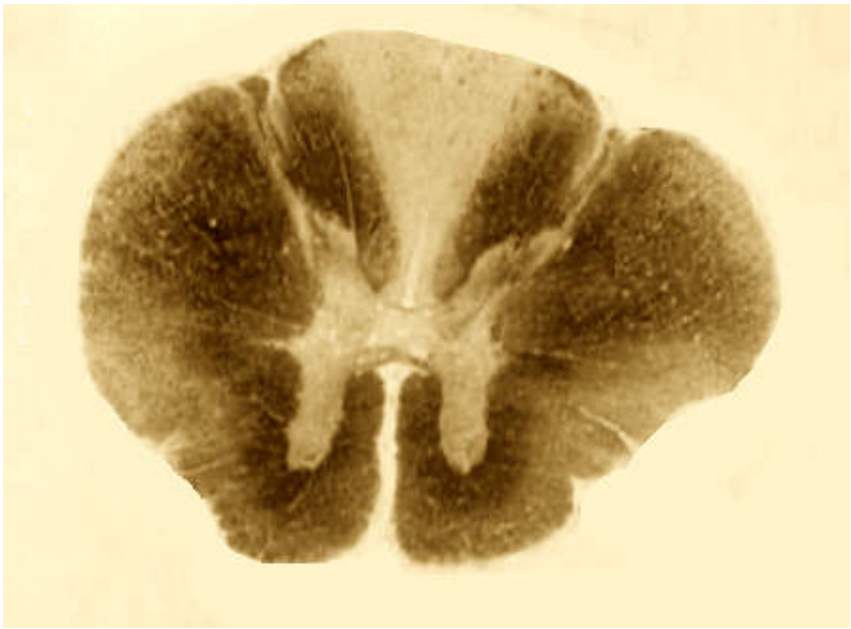


- Choice A = dorsal horn; B = dorsal root; C = dorsal root ganglion; D = lateral horn; E = ventral horn; F = anterior white commissure; G = ventral root; H = spinal nerve.



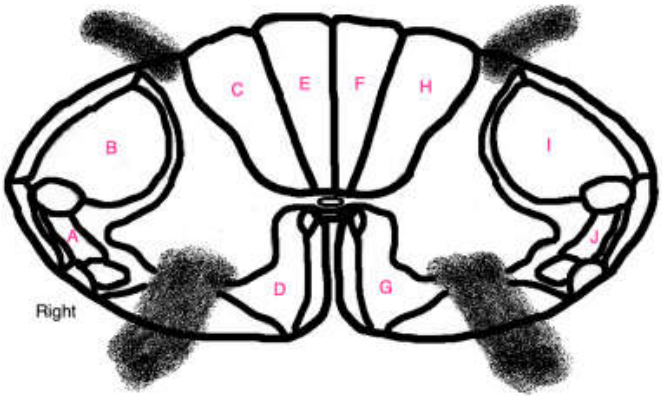
  - A = right corticospinal tract; B = right spinothalamic tract; C = right fasciculus cuneatus; D = right fasciculus gracilis; E = left fasciculus gracilis; F = left fasciculus cuneatus; G = left corticospinal tract; H = left spinothalamic tract.
  - Fasciculus cuneatus → carries vibration/proprioception from the upper limbs + trunk.
  - Fasciculus gracilis → carries vibration/proprioception from the lower limbs.

- 45M + boating accident years prior to death + autopsy is performed in which a myelin stain is used; what was the most likely deficit in this patient?



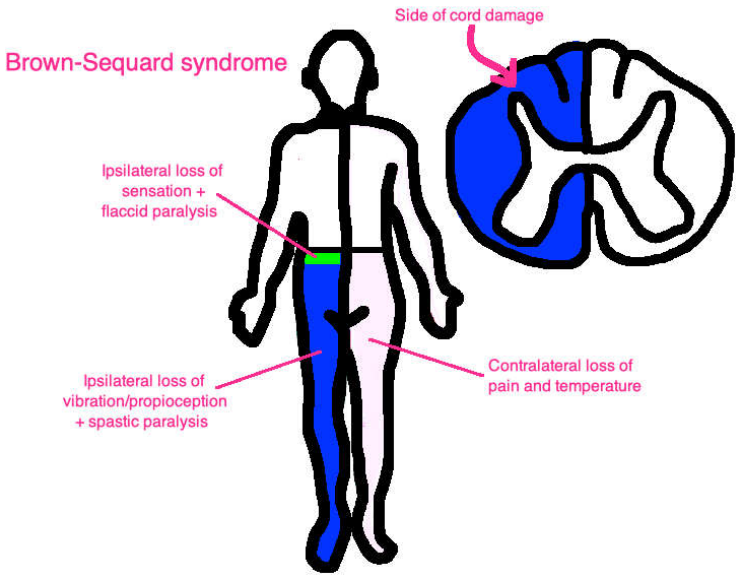
- Answer = loss of vibration/proprioception of the lower limbs → there are bilateral lesions of the fasciculus gracilis of the dorsal columns (only white matter part of the spinal cord not stained black with the myelin stain, meaning it was demyelinated).

- 43F + inability to sense pain/temperature over right lower extremity; which letter on the following spinal cord cross-section best represents the location of her pathology?

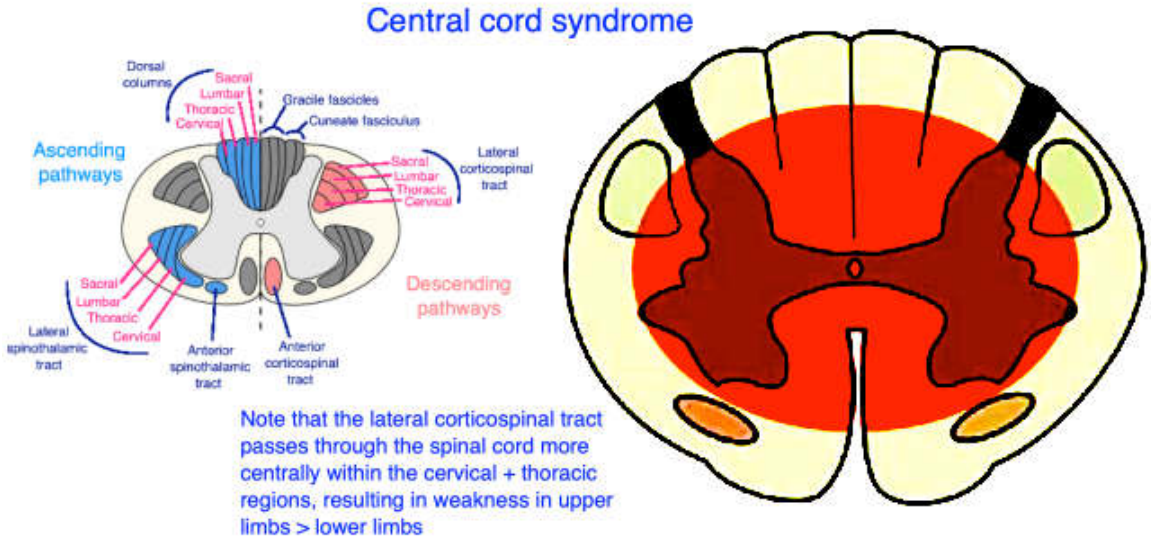


- Answer = left spinothalamic tract (choice J); spinothalamic tract lesions are contralateral; A = right spinothalamic tract; B = right lateral corticospinal tract; C = right fasciculus cuneatus; D = right anterior corticospinal tract; E = right fasciculus gracilis; F = left fasciculus gracilis; G = left anterior corticospinal tract; H = left fasciculus cuneatus; I = left lateral corticospinal tract.

- 24M + viral infection + loss of left leg motor function + loss of left leg vibration/proprioception + loss of right leg pain/temperature sensation; Dx? → answer = Brown-Sequard syndrome → hemisection of the spinal cord; classically stab wound, but can also be autoimmune (i.e., secondary to lupus myelitis or viral infection).

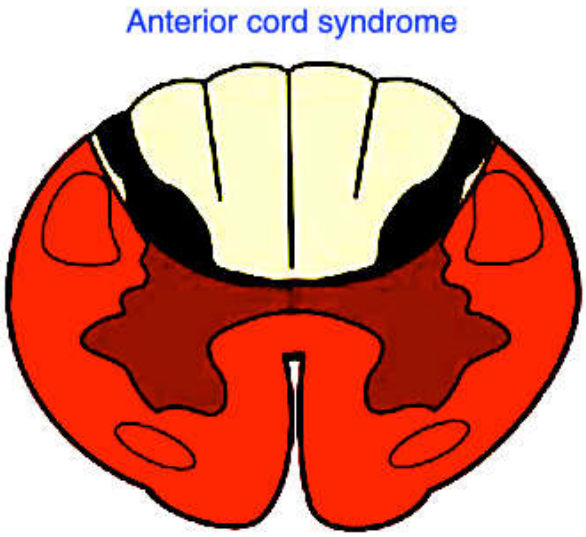


- 67M + long history of cervical spondylosis + whiplash injury after being rear-ended in MVA + weakness in upper limbs > weakness in lower limbs; Dx? → answer = central cord syndrome.



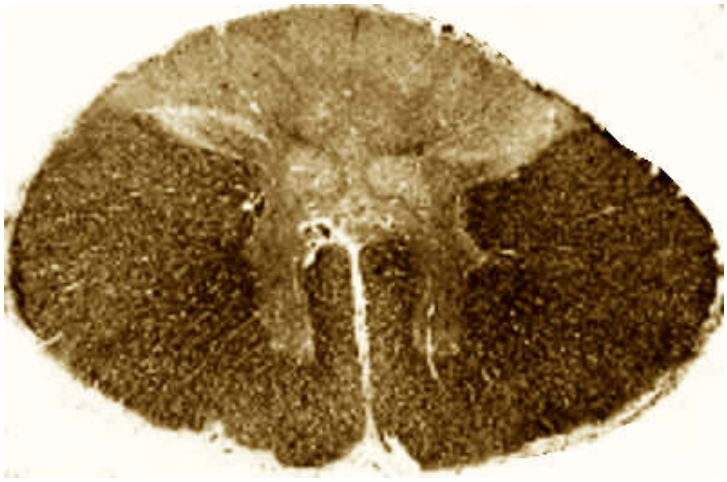
  - Central cord syndrome is the answer when weakness in the upper limbs is greater than weakness in the lower limbs; typically follows hyperextension injury of the neck during whiplash in a rear-end MVA.

- 69M + long Hx of T2DM + bilateral loss of motor function and pain/temperature below level of nipples + intact vibration/proprioception; Dx? → answer = anterior cord syndrome (dorsal columns spared); often due to atherosclerosis + infarction of anterior spinal artery.

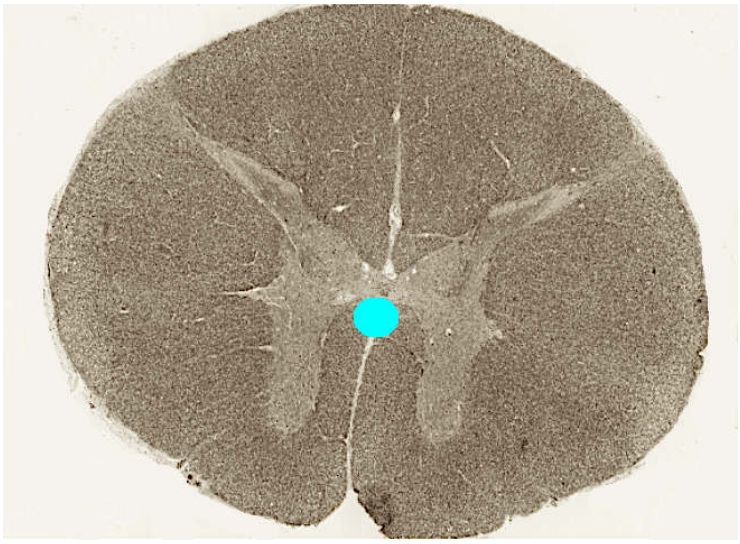


  - Anterior cord syndrome is the answer when there's preservation of vibration/proprioception but loss of motor + pain/temperature, classically in patient with arterial disease.

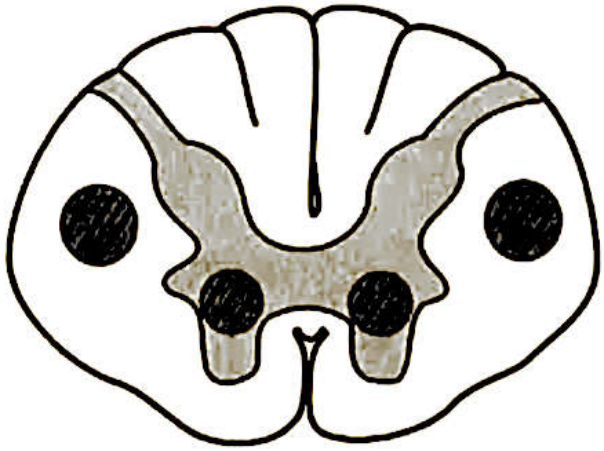
- 23M + presents to the physician with neurologic dysfunction + the following spinal cord cross-section has been stained using a myelin stain; what is the most likely diagnosis?



- Answer = tabes dorsalis → neurosyphilis → loss of dorsal columns → dorsal columns are the only part not stained, reflective of demyelination → isolated loss of vibration/proprioception; motor function and pain/temperature sensation intact.
- 31M + motorcycle accident + presents with spinal lesion affecting the following area on the cross-section; what type of neurologic dysfunction is most likely in this patient?

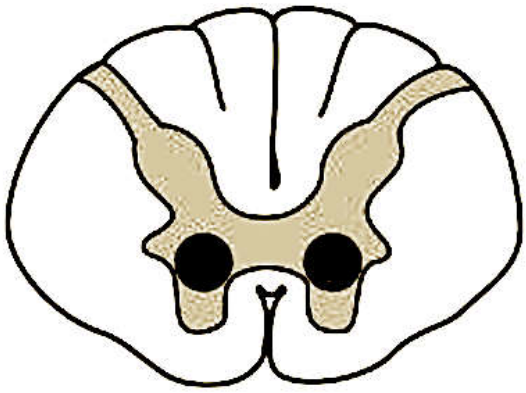


  - Answer = bilateral loss of pain and temperature; diagnosis is syringomyelia (lesion of anterior white commissure); can be due to trauma, infections, and classically Chiari I.
- 54M + clonus and Babinski reflex in left foot + fasciculations in right hand + brisk reflexes + elevated serum creatine kinase (CK); Dx? → answer = amyotrophic lateral sclerosis (ALS; Lou-Gehrig disease); two components to the Dx:



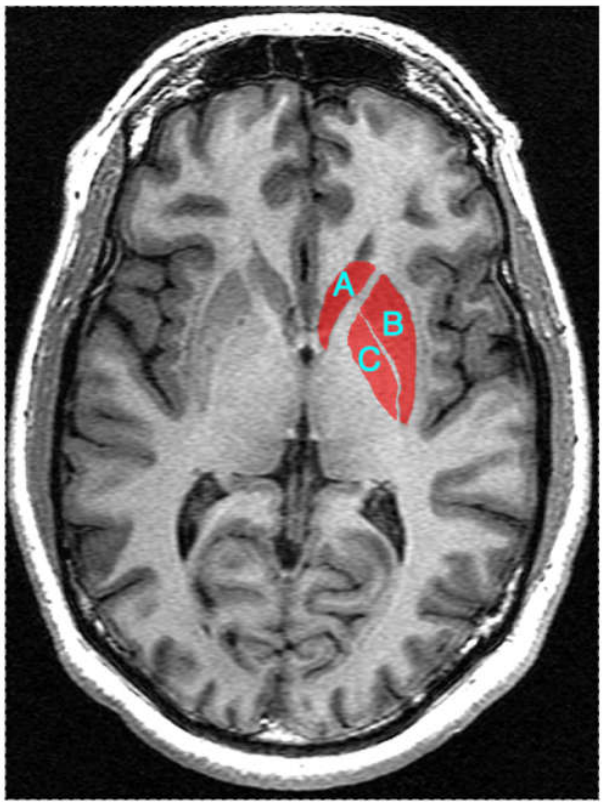
- In ALS, the lateral corticospinal tracts (UMN) and anterior horns (LMN) degenerate.
- 1) Vignette will always have combination of upper motor neuron (UMN) and lower motor neuron (LMN) findings. Fasciculations, decreased reflexes, decreased tone, and muscle atrophy are classic for LMN. Babinski reflex, clonus, increased (brisk) reflexes, increased tone are classic for UMN. Serum CK can be elevated due to increased tone.
- 2) Must have **no** sensory findings. If any sensory abnormalities (i.e., paresthesias, numbness, etc.) are present in the vignette, ALS is the wrong answer.
- USMLE will often give vignette of ALS and then the answer is simply "motor neurons."
- HY for 2CK in particular to know answer for Dx = "electromyography and nerve conduction studies."

- 55M + immigrant from Albania + presents to GP for first time + physical exam shows a left leg with remarkably decreased muscle bulk and diminished motor function; Dx? → answer = polio → loss of anterior horns of spinal cord.



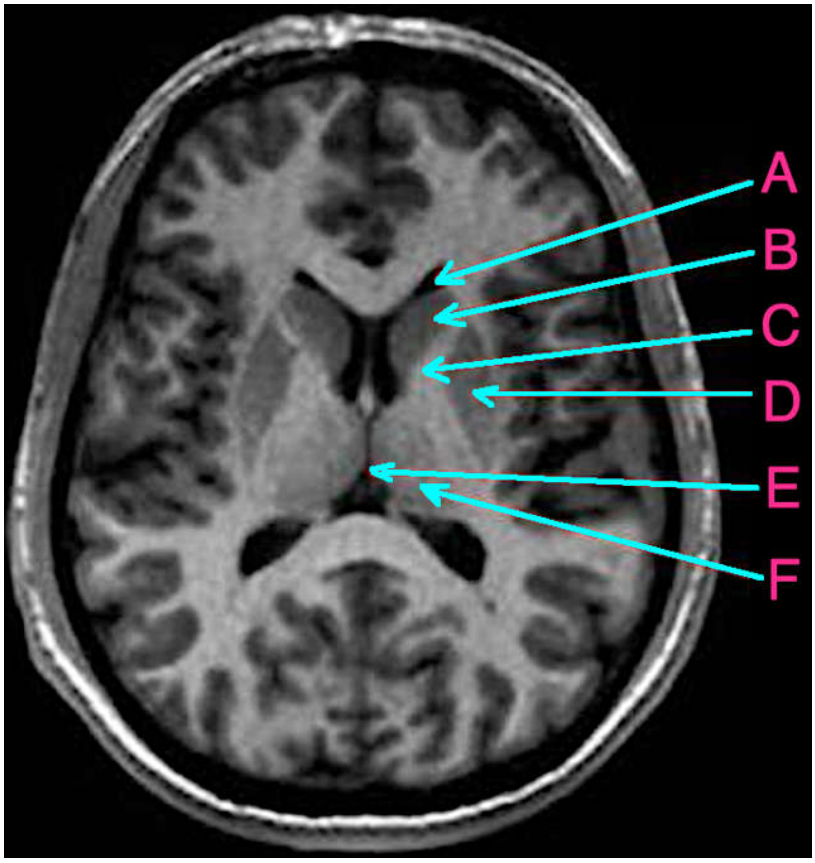
- Anterior horns = the answer for the location of the neurologic dysfunction seen in poliomyelitis.
- Anterior horns also = the answer for progressive infantile spinal muscular atrophy (Werdnig-Hoffman syndrome); this Dx is on the pediatric 2CK shelf → presents with profound hypotonia, absent reflexes, and **tongue fasciculations.**

- 23F + unilateral restring tremor + increased hepatic ALT and AST + low hematocrit; which letter on the following MRI correctly labels this patient's putamen?



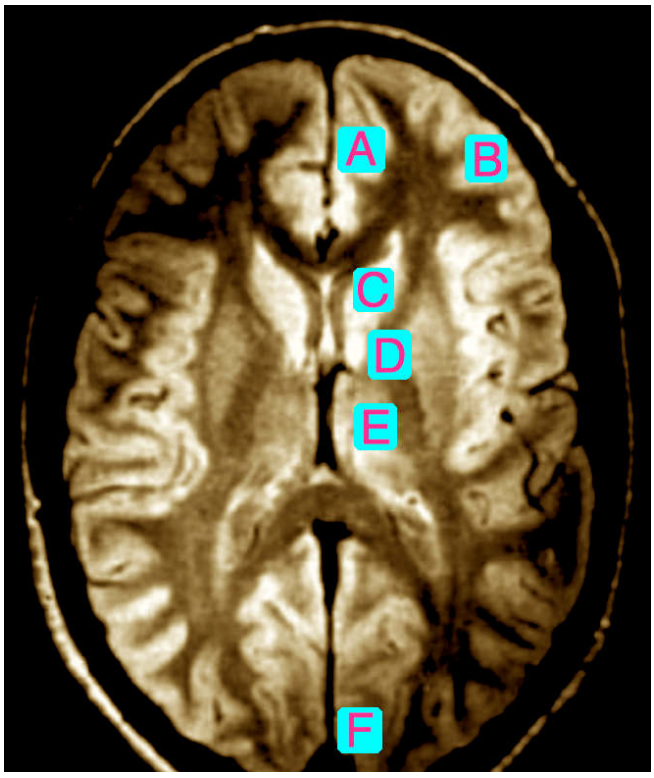
- A = caudate nucleus; B = putamen; C = globus pallidus.
- Parkinsonism in a young patient is Wilson disease until proven otherwise; copper deposition can occur anywhere in the basal ganglia; the putamen is described as a classic location.

- 41M + one-year Hx of jerking movements of extremities + 6-month Hx of worsening dementia + father had similar presentation and died at age 56; where on the following MRI is the location of this patient's pathology?



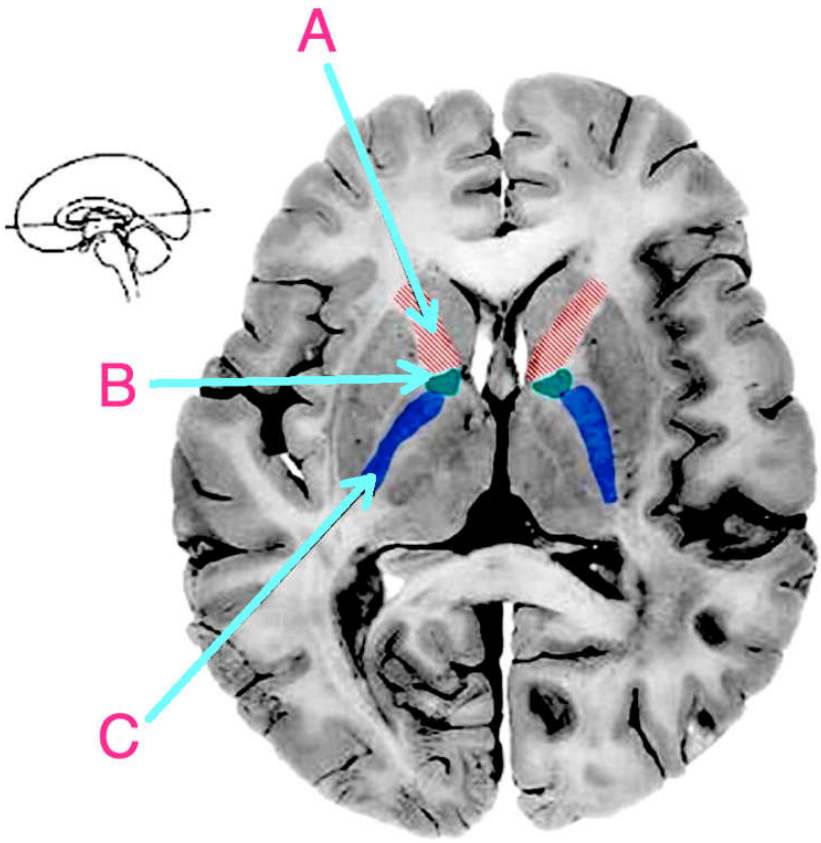
- The answer is the caudate nucleus, which is choice B. Choice A = anterior horn of lateral ventricle; C = internal capsule; D = putamen; E = third ventricle; F = thalamus.

- Identify:



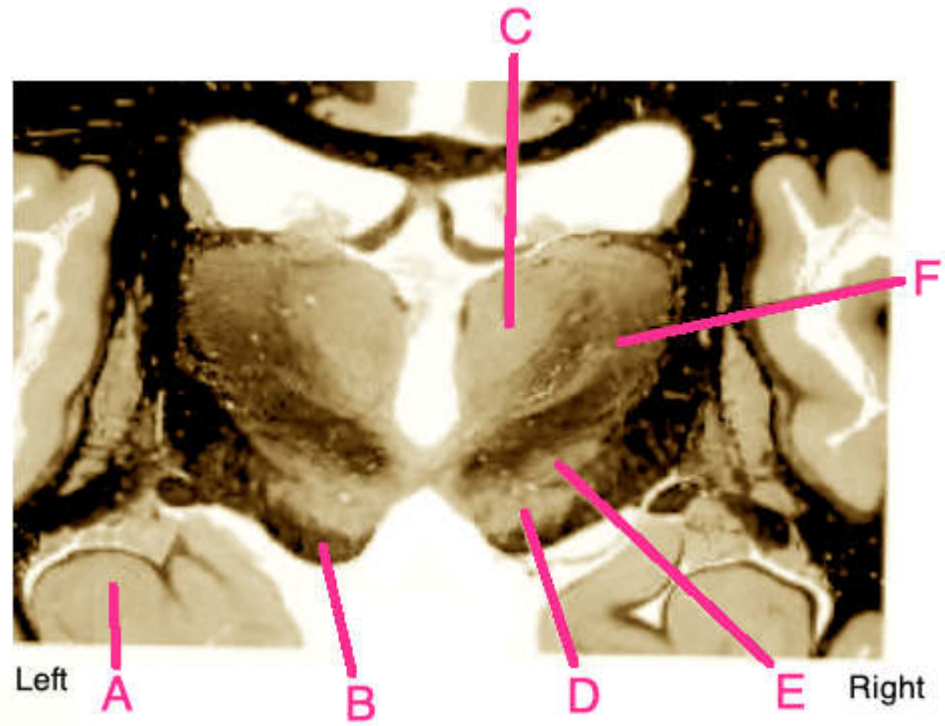
- A = frontal lobe (medial frontal gyrus); B = frontal lobe (middle frontal gyrus); C = caudate nucleus; D = internal capsule; E = thalamus; F = occipital lobe (occipital pole).

- 78M + long Hx of HTN + sudden paresis of left side of his body; which letter on the following specimen is the most likely location of his stroke?



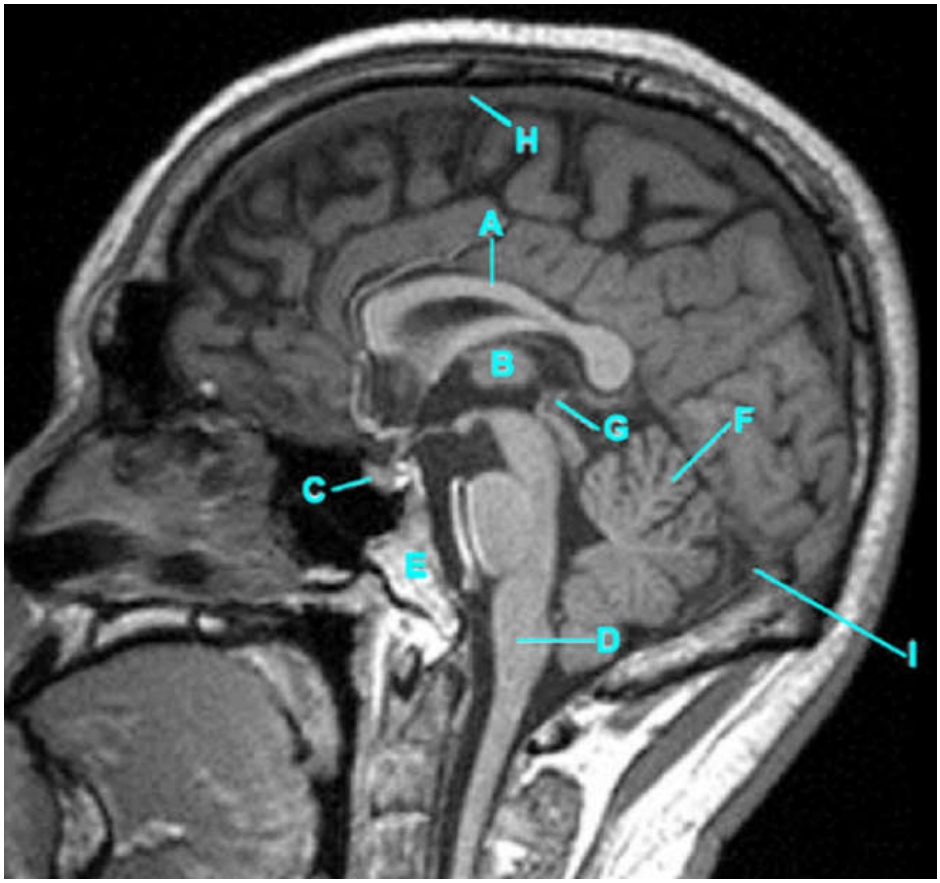
- The answer is C (posterior limb of internal capsule). A = anterior limb of internal capsule; B = genu of internal capsule.
- USMLE wants you to know that the lenticulostriate vessels supply the internal capsule. Chronic hypertension leads to lipohyalinosis and compromise of these vessels. Posterior limb infarcts can sometimes result in pure motor strokes with contralateral hemiparesis/-plegia.
- Lesions of posterior limb of internal capsule are sometimes described as causing “pure motor” dysfunction.

- 63M + longstanding Hx of HTN + acute-onset flailing movements of left arm; where on the following cross-section of the corpus striatum is the most likely location of the patient’s pathology?



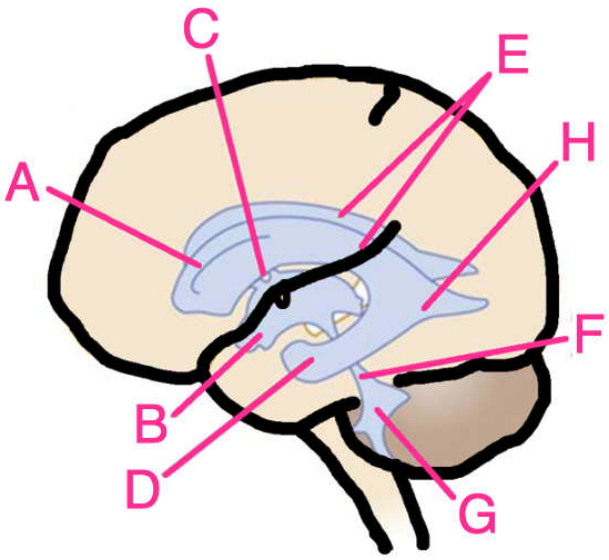
Answer = subthalamic nucleus (choice E); A→ hippocampus; B = crus cerebri; C = ventral medial nucleus of thalamus; D = substantia nigra; E = subthalamic nucleus; F = ventral lateral nucleus of thalamus.

- Identify:



- **A = corpus collosum; B = thalamus; C = pituitary gland;** D = medulla; E = clivus; F = cerebellum; **G = pineal gland;** H = superior sagittal sinus; I = torcular herophili.

- 32M + presents with hydrocephalus + MRI shows colloid cyst most likely located where on the following illustration?



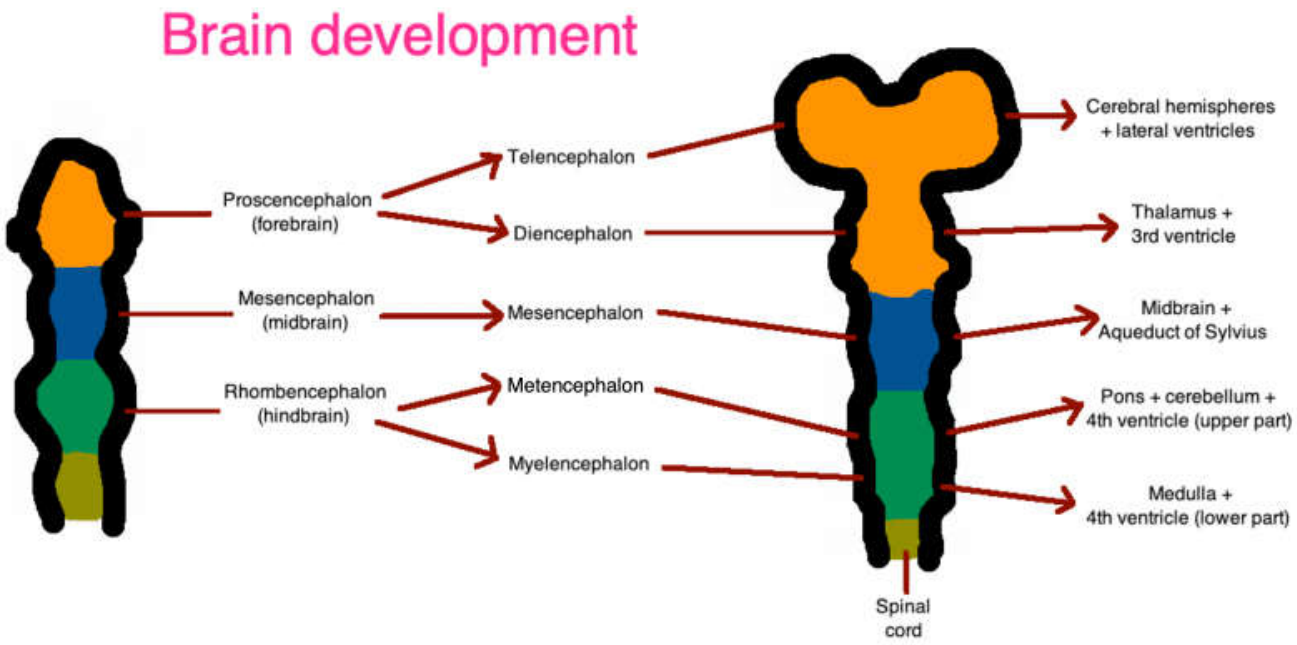
- Answer = choice C (foramen of Monro); colloid cysts are almost always found originating from the roof of the anterior third ventricle, just posterior the foramen of Monro. Weirdly specific, but just be aware. A = frontal horn; B = third ventricle; D = temporal horn; E = lateral ventricles; F = aqueduct of Sylvius; G = fourth ventricle; H = occipital horn.
- Cerebral aqueduct of Sylvius → connects third and fourth ventricles.
- Foramen of Monro → connects the third ventricle to both lateral ventricles.

- Identify:

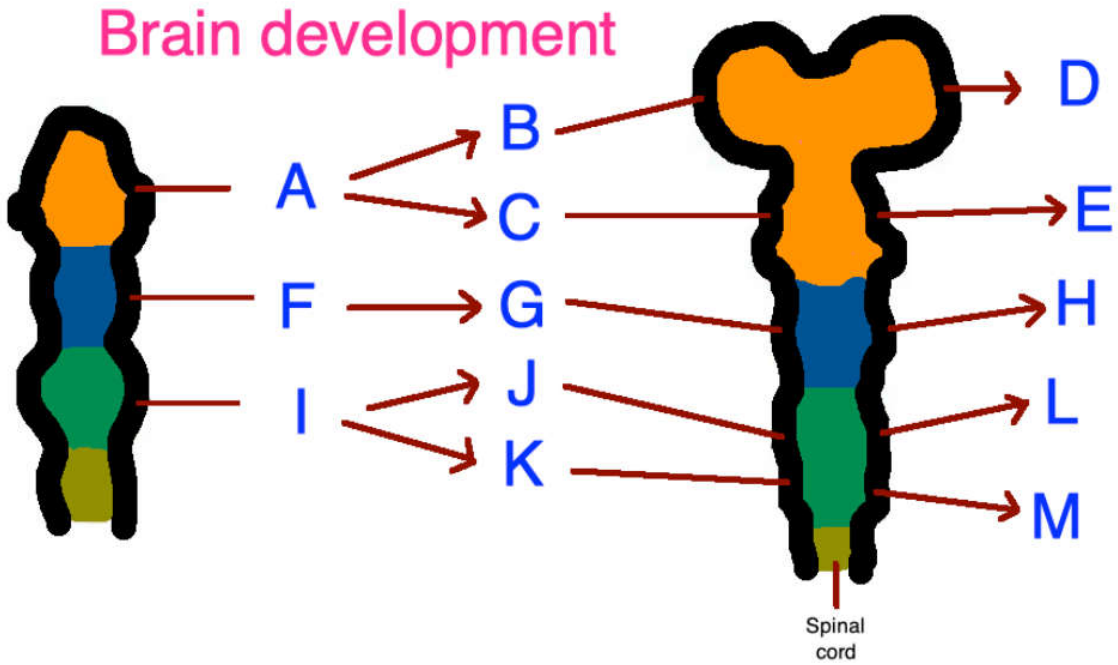


- A = lateral ventricles; **B = colloid cyst of roof of the third ventricle;** C = third ventricle.

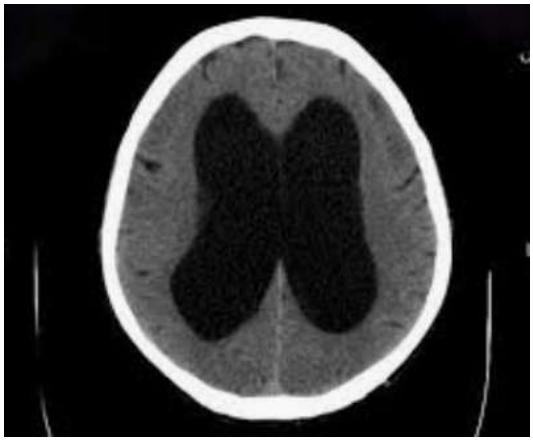
"How would the brain development stuff like diencephalon, mesencephalon, etc., be asked?" → First learn the following diagram:



- Identify:

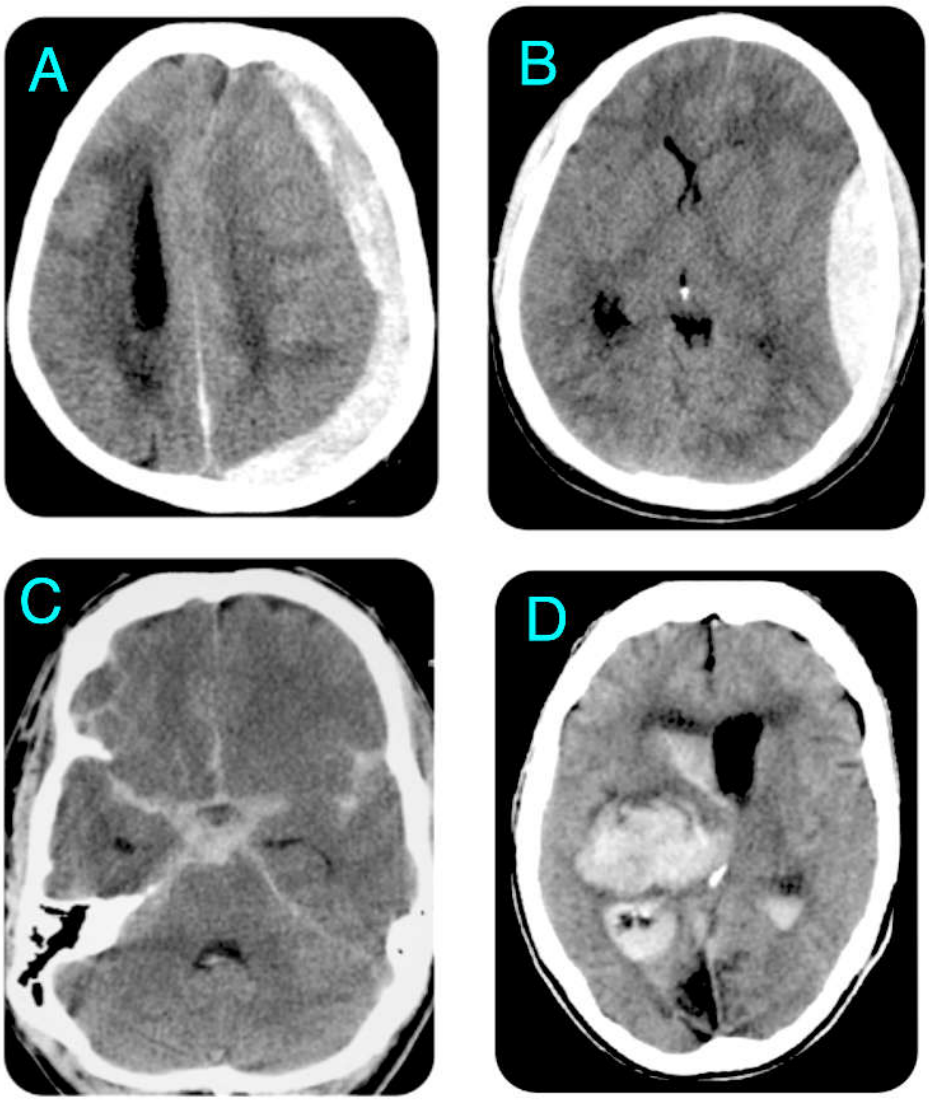


- 72M + unsteady gait + personality change + urinary incontinence + Parkinsonism; the enlarged structures labeled on the following CT are derived from which component of brain development (i.e., prosencephalon, diencephalon, etc.)?



- Answer = telencephalon of the prosencephalon → dilated lateral ventricles are seen in normal pressure hydrocephalus (wet, wobbly, wacky +/- Parkinsonism).

- 84M + one-year Hx of progressive cognitive decline + no Hx of HTN, CVD, or atrial fibrillation + experiences a stroke and dies; which of the following head CTs might best reflect this patient?



- Answer = intracerebral (intraparenchymal) hemorrhage (choice D); Dx is amyloid angiopathy resulting in hemorrhagic stroke in Alzheimer disease; A = subdural hematoma; B = epidural hematoma; C = subarachnoid hemorrhage.
- Subdural hematoma (choice A) → rupture of bridging (superior cerebral) veins; **no lucid interval (i.e., patient did not lose consciousness);** crescent-shaped; can be slowly accumulating and heterogenous in appearance due to combination of dried + fresh blood; increased risk in elderly/dementia and alcoholics; answer in acceleration-deceleration injuries (shaken baby syndrome; motor vehicle accidents); NBME wants craniotomy as the answer for Tx (no evidence for mere observation as the answer on any of the USMLE material); if the patient has decreased Glasgow score, do "intubation + hyperventilation" as answer (decreased CO2 → decreased cerebral perfusion → will reduce intracranial pressure) before craniotomy.
- Epidural (choice B) → rupture of middle meningeal artery (branch of external carotid); **lucid interval (i.e., patient lost consciousness** then returned to normal → patient goes home thinking he/she is okay and then dies)**;** lens (biconvex) shape; fast accumulating; if low GCS, do "intubation + hyperventilation," *then* do craniotomy. If normal GCS, just do craniotomy.
- Subarachnoid (choice C) → rupture of anterior communicating artery (ACoM) or posterior communicating artery (PCoM) saccular/berry aneurysm; ACoM > PCoM in terms of location; patient often reports "worst headache of his/her life"; **PCoM aneurysm classically associated with ipsilateral blown pupil (mydriasis)** due to CN III impingement; HTN is most common risk factor in population and accounts for most cases; Ehlers-Danlos and autosomal dominant polycystic kidney disease (ADPKD) are HY non-HTN-associated specific causes of saccular aneurysms. Patient with HTN still more likely to die from myocardial infarction (HTN → atherosclerosis) than saccular aneurysm (asked on NBME).
- Intracerebral (intraparenchymal) hemorrhage → associated with Alzheimer (amyloid angiopathy especially the answer in Alzheimer patients with recurrent hemorrhagic stroke); also associated with chronic HTN and Charcot-Bouchard aneurysms of lenticulostriate arteries; can be associated with brain cancer as well (e.g., glioblastoma multiforme).

- 17M + motorcycle accident + severe cognitive impairment two years following accident; based on the following MRI, what is the most likely diagnosis?



  - Answer = diffuse axonal injury; caused by shearing forces from rapid-deceleration injury.



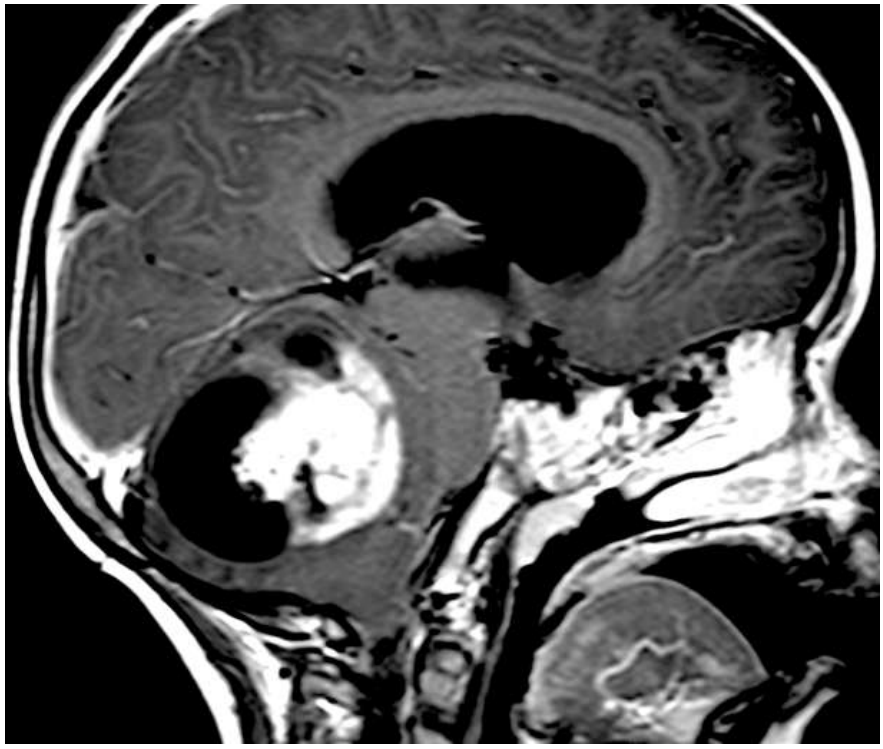
  - Note the irregular areas of hyperintensity on diffusion-weighted MRI.

- 44F + SLE + the following CT of head shows ring-enhancing lesion; what's the diagnosis?



  - Answer = primary CNS lymphoma; the wrong answer is Toxoplasmosis. Ring-enhancing lesions can refer to abscesses, malignancy, Toxo, and neurocysticerosis.
  - Autoimmune diseases (e.g., Hashimoto, SLE, etc.) and immunosuppression (i.e., AIDS) carry increased risk of primary CNS lymphoma.

- 3M + failure to thrive + seizure; based on the following MRI, what's the most likely diagnosis?



  - Answer = pilocytic astrocytoma. Notice the large **solid and cystic** lesion. The solid component is white (hyperintense) on the MRI; the cystic component is black (hypointense). This is a HY image for Step 1.

- 54M + 3-month Hx of cognitive disturbance + presents today with seizure; based on the following head CT, what's the most likely diagnosis?



  - Answer = glioblastoma multiforme (butterfly glioma); notice the heterogeneity due to areas of necrosis and hemorrhage; it need not cross the midline, but when it does it is sometimes referred to as "butterfly glioma."

- 16M + developed cataracts at young age + six-month history of left-sided tinnitus and hearing loss; if his diagnosis is the same as that seen in the following gross specimen, what is his most likely underlying genetic condition?

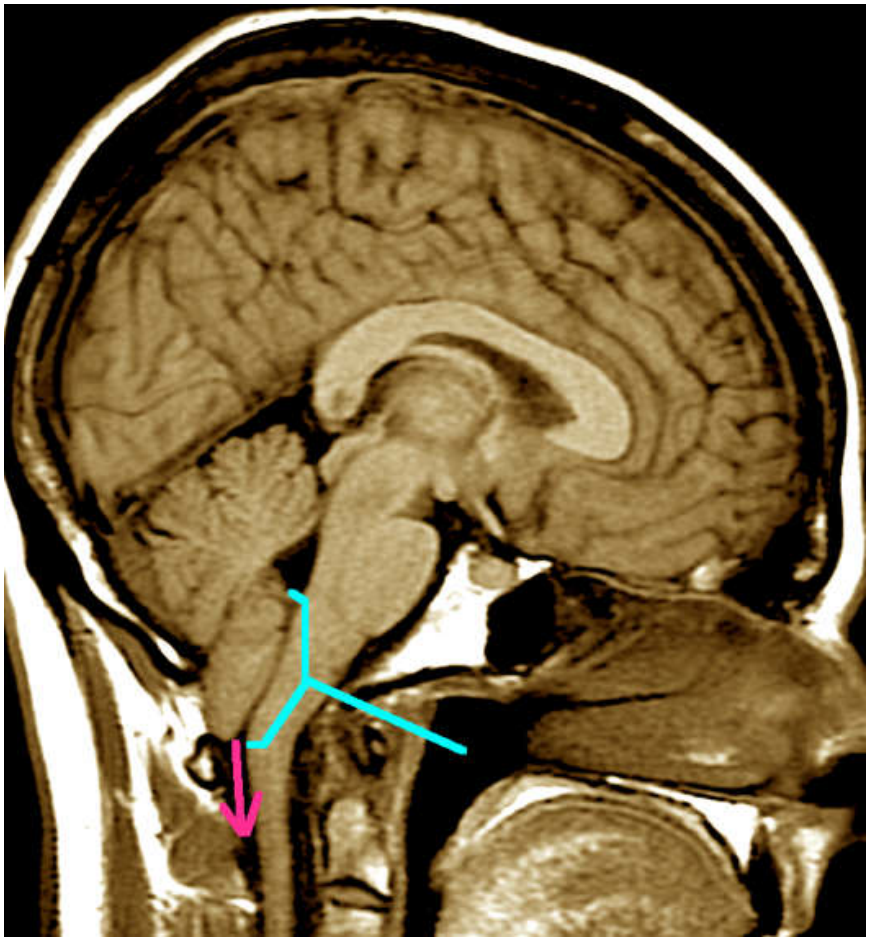


- Dx is acoustic schwannoma occurring at the cerebellopontine junction; in a young patient, neurofibromatosis type II (NF2) is likely etiology; can progress to be bilateral; if elderly patient, usually sporadic and unilateral.

- 56M + nine-year Hx of progressive right-sided hearing loss; MRI shows large mass at the cerebellopontine junction; the mass most likely arises from which nerve and cell type? → answer = vestibulocochlear nerve and neural crest cells.
- Looking at the following MRI, what's the diagnosis?

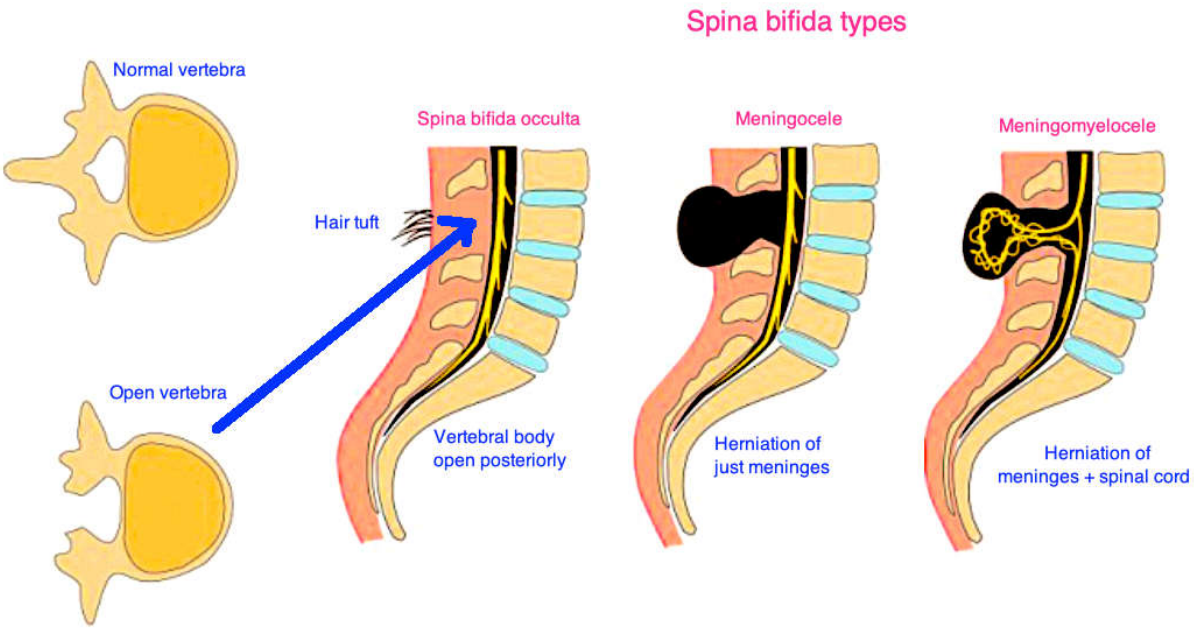


- Answer = Chiari malformation, characterized by inferior herniation of the cerebellar tonsils.



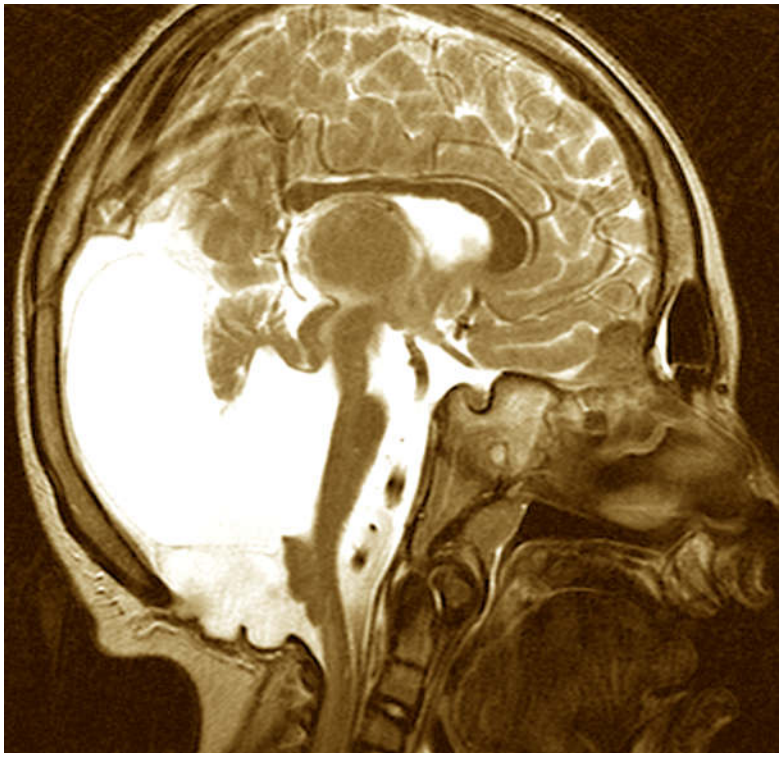
- Note the inferior herniation of the cerebellar tonsils; Chiari type I is characterized by mere herniation and is the answer if the vignette gives you a **teenager or young adult;** it can also be associated with syringomyelia (syrinx of CSF in the spinal cord). Chiari II (Arnold-Chiari malformation) is more severe and is the answer if the vignette mentions a **child** who not only has cerebellar herniation but also lumbosacral myelomeningocele of the spinal cord. In other words:
- Teenager or young adult + syrinx of spinal cord → answer = Chiari I.
- Child + lumbosacral meningomyelocele → answer = Chiari II (Arnold-Chiari malformation).

- Learn the following herniation types:



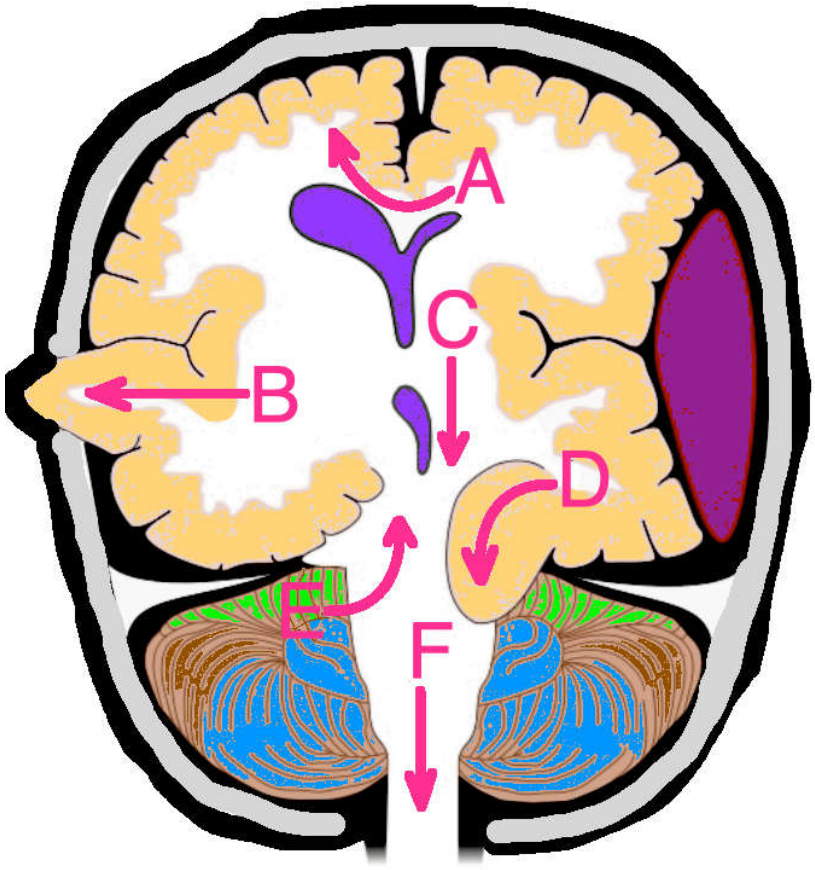
- HY point is that neural tube defects (spina bifida) occur **between 3-4 weeks gestation.** The pregnant woman needs adequate folic acid (vitamin B9) prior to and during early pregnancy for prevention. Antiepileptics (i.e., particularly valproic acid, but also phenytoin and carbamazepine) are also associated with neural tube defects.
- Neural tube defects lead to increased AFP and acetylcholinesterase in CSF.

- Newborn + myelomeningocele; what is the affected developmental process? → answer = "closure of caudal neuropore."
- 23F at 20 weeks gestation + ultrasound shows fetus with anencephaly; what is the mechanism for this condition? → "failure of closure of rostral neuropore."
- Neonate + tuft of hair on midline in lumbar region + palpation of region shows absence of spinous process; what is the mechanism for this condition? → answer = "failure of fusion of the sclerotomes."
- Looking at the following MRI, what's the diagnosis?



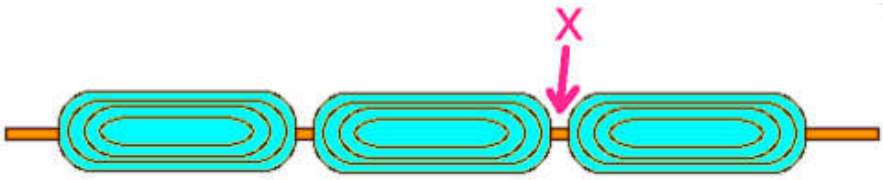
- Answer = Dandy-Walker malformation; all you need to know: this is the answer if the vignette describes absent cerebellar vermis + cystic dilation of the fourth ventricle.

- Identify the following brain herniation types:



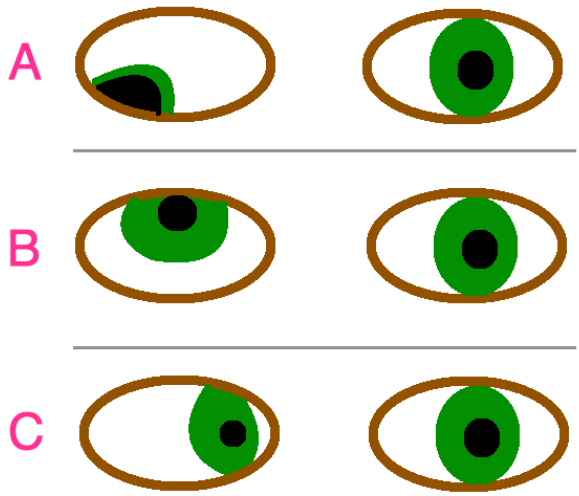
- A = cingulate (subfalcine) herniation; B = transcalvarial herniation; C = central transtentorial herniation; D = uncal herniation; E = upward transtentorial hernation; F = tonsillar herniation.

- The area labeled X on the following diagram notably contains a high density of which kind of channel(s) (i.e., Na, Ca, K, etc.)?



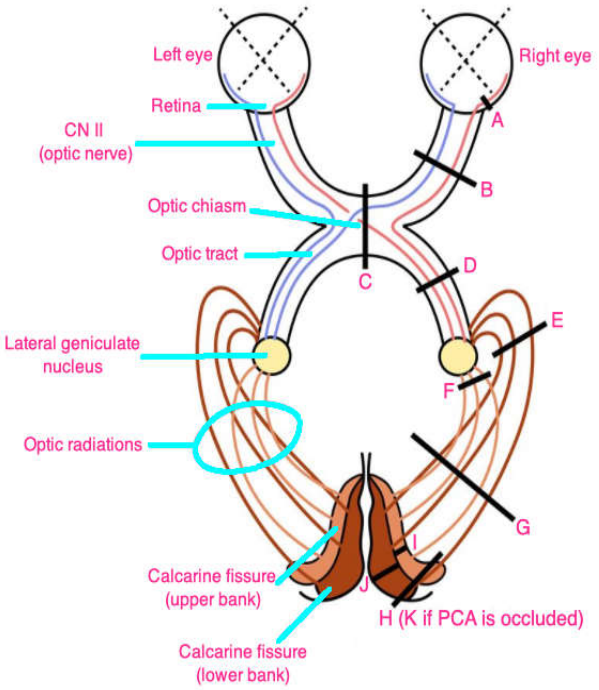
- Area labeled X = node of Ranvier; answer = contains high concentrations of sodium channels.

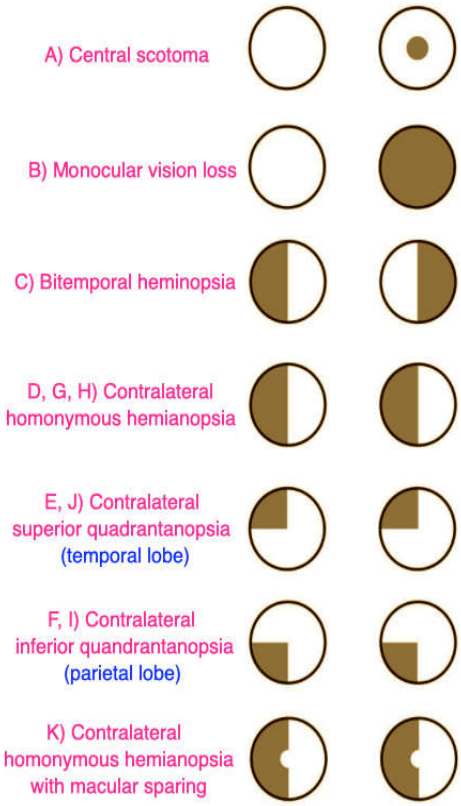
- Identify the following cranial nerve lesions:



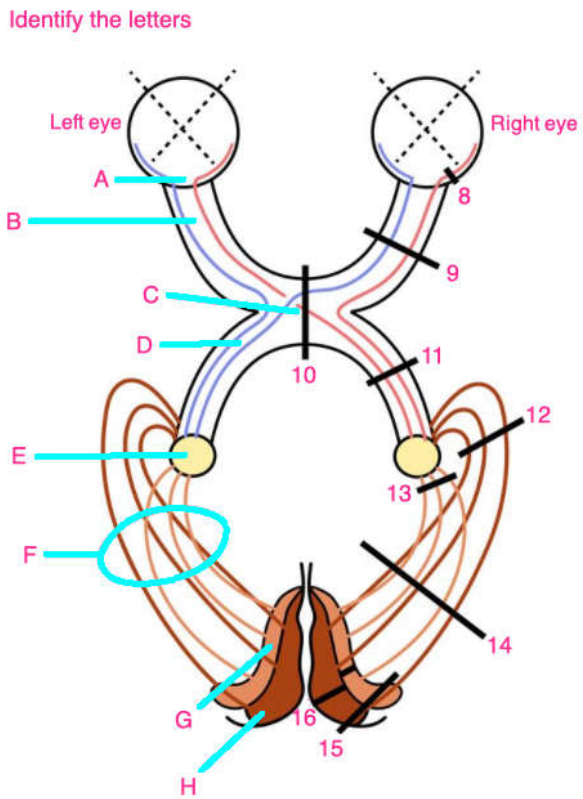
- A = lesion of CN III (oculomotor) → presents as down and out; B = CN IV lesion (trochlear) → presents as upward gaze + patient will tilt head toward unaffected side; C = CN VI (abducens) → medially directed eye.
- PCoM aneurysmal compression of CN III affects superficial parasympathetic efferents leading to ipsilateral mydriasis and loss of accommodation; diabetes may affect deep extraocular muscle efferents leading to partial ptosis and down-and-out pupil.

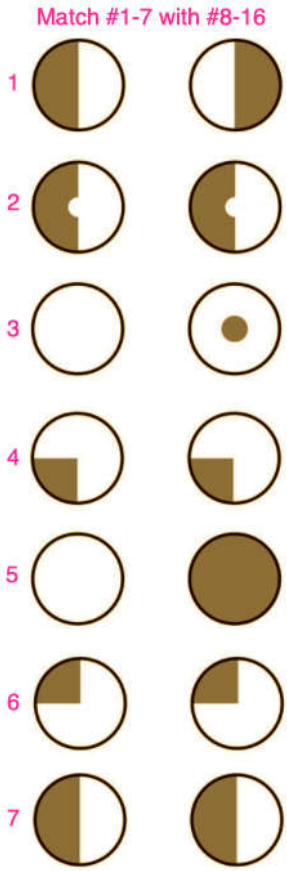
- The visual pathway is important for Step 1:





- After you memorize the above, now try this:

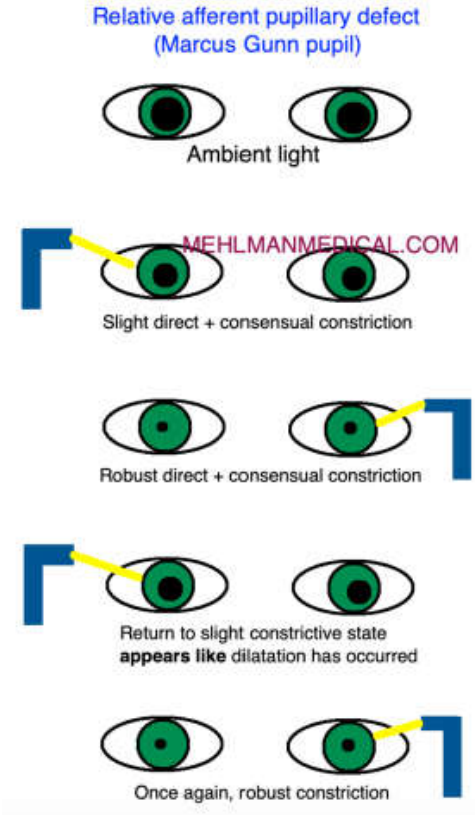




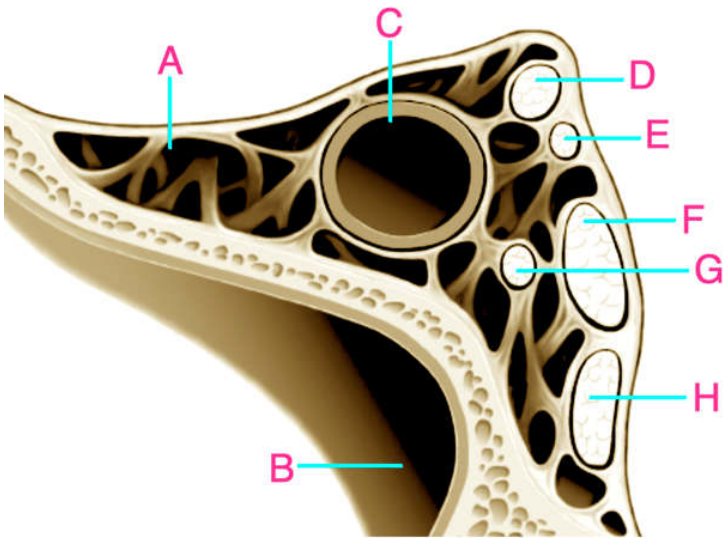
- 31F + Hx of multiple sclerosis + presents with the following visual problem; where is the location of her pathology?



  - Answer = right parietal lobe (contralateral inferior quadrantanopsia).
- 45F + lung cancer + 2-week Hx of progressive loss of vision of the left visual field of both eyes; if MRI of the brain is performed, metastases might be seen where? → answer = right occipital Lobe; Dx is homonymous hemianopia.
- "Can you explain that Marcus-Gunn pupil and optic neuritis stuff?" → MG pupil caused by CN II (optic nerve) lesion, classically as a result of optic neuritis in flare of multiple sclerosis; because of the CN II lesion, the intensity of the light is under-sensed by that eye (afferent signal), so the resultant efferent CN III (oculomotor) parasympathetic output (back to the eyes for pupillary constriction) is attenuated; this means when you shine a light in the affected eye, both pupils constrict less in comparison to when the light is shone in the unaffected eye; this **gives the mere impression** that the pupils dilate when light is shone in the affected eye. Still confusing? Take a look at the following illustration (assuming a lesion of the patient's right eye [left side of illustration]):

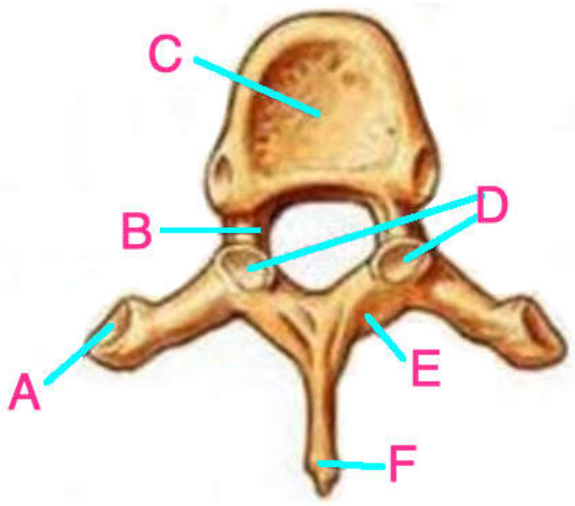


- 45M + presents with the left pupil larger than the right pupil after undergoing cranial surgery; which letter on the following image best reflects the location of his pathology?

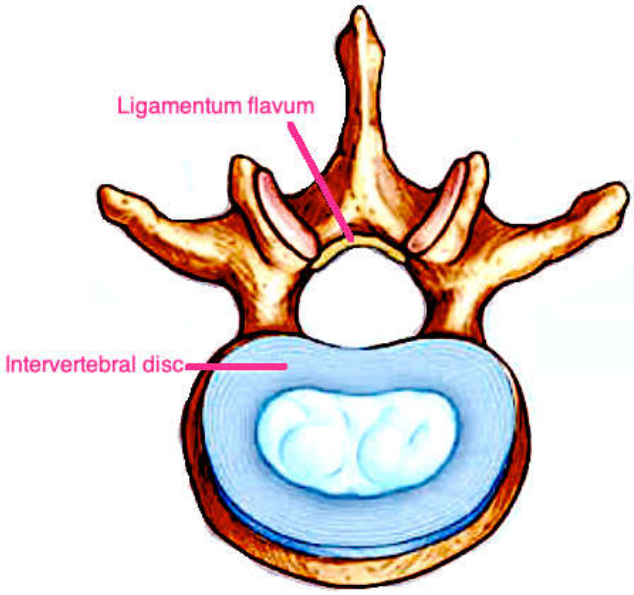


  - Answer = D (CN III; oculomotor nerve), lesion can cause down and out pupil and/or mydriasis; this illustration of the cavernous sinus is exceedingly HY for Step 1. A = cavernous sinus; B = sphenoid sinus; C = internal carotid artery; E = trochlear nerve (CN IV); F = ophthalmic branch of trigeminal nerve (V1); G = abducens nerve (CN VI); H = maxillary branch of trigeminal nerve (V2).
- 34M + sinus infection one week ago treated with amoxicillin/clavulanate + now presents with diplopia, ophthalmoplegia, and headache + has fever of 103F; Dx? → answer = cavernous sinus thrombosis; sinus infections can ascend to cavernous sinus via the valveless facial veins; the sequence of sinus infection followed by fever, diplopia, and ophthalmoplegia is classic for cavernous sinus thrombosis; give IV antibiotics.
- 41F + new-onset neurologic dysfunction following surgery of her sphenoid sinus; what additional finding is likely present on the examination? → inability to abduct eye (abducens nerve injury); CN VI is most medial / closest to the sphenoid sinus and is susceptible to injury.
- 49M + prolactinoma + bitemporal hemianopsia; Q asks what other eye finding might be present; answer = inability to abduct eye (abducens nerve injury); a pituitary tumor within the sella turcica expanding laterally can cause CN VI dysfunction; superior extension of pituitary tumor causes classic bitemporal heminanopsia due to impingement on optic chiasm.

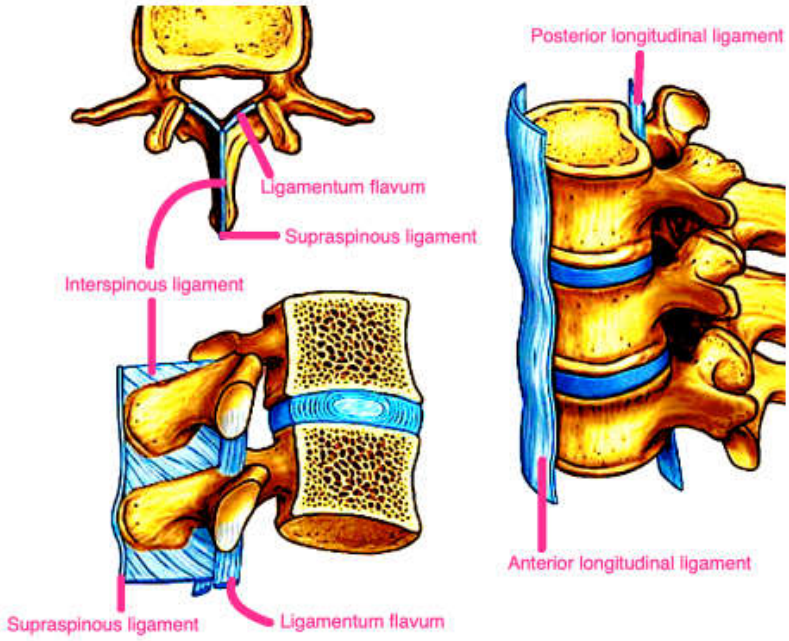
- Infection in the face may spread to cavernous sinus via valveless facial venous system (via Sup. & Inf. Ophthalmic veins) leading to cavernous sinus thrombosis where Abducent nerve is the most susceptible nerve to injury.
- 22M + needs to undergo laminectomy; where on the following illustration of a vertebra is the most appropriate location for surgical entrance to the neural canal?



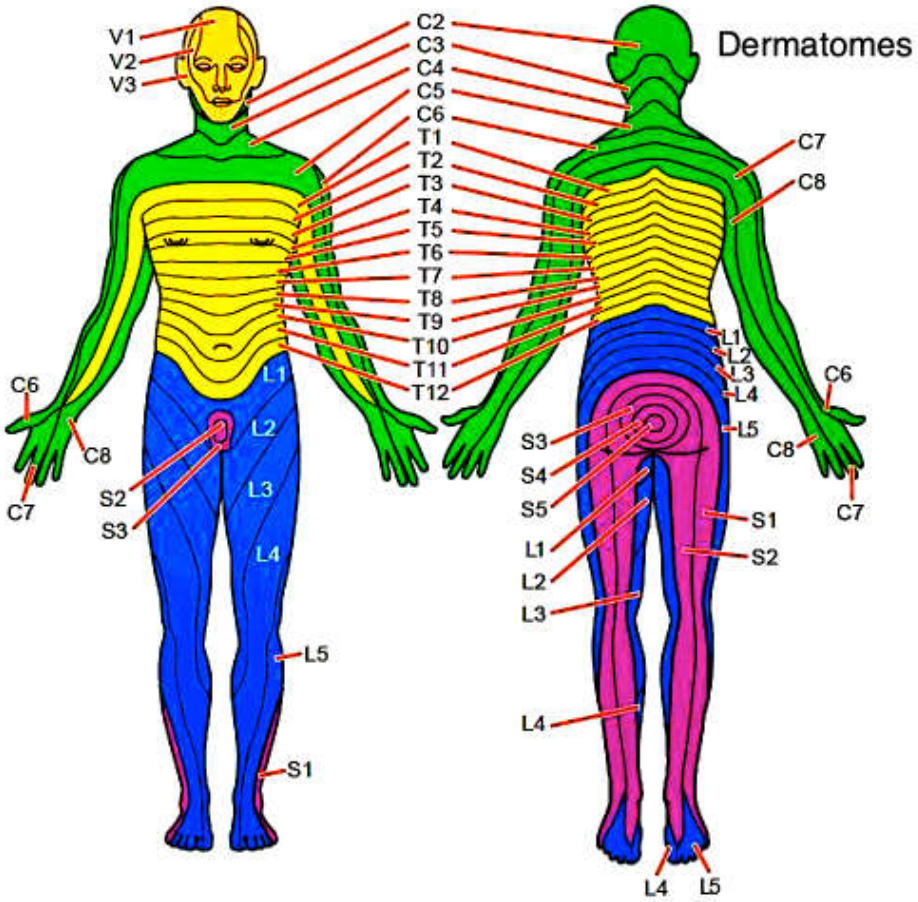
  - Answer = E (lamina); A = transverse process; B = pedicle; C = vertebral body; D = facet joints; F = spinous process.
  - Should be noted as a tangential factoid that the vertebral arteries (first branch of subclavians) ascend bilaterally through the transverse foramina of the cervical vertebrae.
- 17M + undergoing spinal surgery + if needle is inserted posteriorly between L3 and L4 vertebral arches, what ligament must be traversed to reach the spinal canal? Answer = ligamentum Flavum.



- 24M + lumbar disc herniation; which spinal ligament is pushed into the spinal root nerves? → answer = posterior longitudinal ligament.



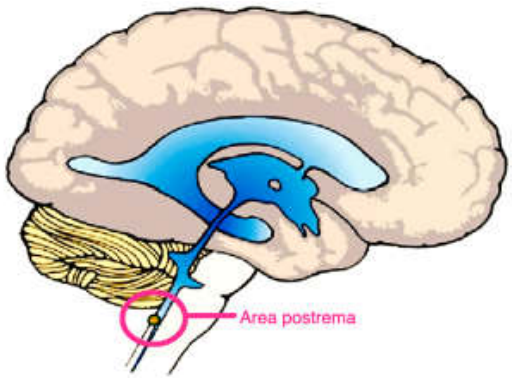
- 63M + pain radiating down distal anterior thigh, knee, medial leg, and medial foot; compression of nerve root in which intervertebral foramina is most likely the cause of her symptoms? → answer = L3-L4.



  - HY dermatomes and reflexes: C5 = biceps jerk; C7 = triceps reflex; T4 = nipples; T10 = umbilicus; L1 = groin; L3/4 = knee jerk; S1 = ankle jerk + sole of foot.

- 32F + gestational pemphigoid with itchy vesicles around the umbilicus; her pain is sensed by which dermatome? → answer = T10.
- 52M + sudden loss of vision in right eye + no pain + central scotoma of visual field + fundoscopy shows pale opaque fundus and bright red fovea centralis; 8 months later, if right eye is illuminated, what reaction is most likely to occur in left pupil? → answer = "no constriction because retinal ganglion cells in right eye have been destroyed"; normal loop should be CN II of right eye signaling (afferent) to midbrain, with CN III relaying efferent parasympathetic activity back to both eyes to cause both direct and consensual pupillary constriction; in this case, since CN II on the right is disrupted, we would get neither a direct or consensual pupillary constriction in response to light shone in the affected eye; Dx is central retinal artery occlusion leading to cherry red spot on macula; etiology either carotid plaque launching off in setting of HTN or mural thromboembolus secondary to AF; patient experiences permanent painless loss of vision.
- 42F + complains of diplopia + can read books normally + decreased adduction in left eye when gazing to the right; what is the likely location of the demyelinating plaque causing these symptoms? → answer = left MLF (internuclear ophthalmoplegia).
- 68M + transient ischemic attack + new-onset anterograde Amnesia; what is the most likely affected location? → answer = hippocampus, which is known to be very sensitive to ischemic insult; it is also an answer sometimes for Alzheimer (progressive shrinking of hippocampus and nucleus Basalis of Maynert).
- 34F + left eye pain + blurry vision + progressive weakness of right arm and both legs + fundoscopy shows optic disc swelling + MRI shows multiple T2-weighted bright signal abnormalities in white matter; what cell is attacked by the immune system in this disease? → answer = oligodendrocytes; Dx is multiple Sclerosis; mechanism is T cell-mediated destruction of oligodendrocytes in CNS (in contrast, Schwann cell destruction in PNS is Guillain-Barre); oligodendrocytes synthesize myelin in the CNS; Schwann cells synthesize myelin in the PNS (including cranial nerves, which are PNS).

- 81M + stroke + which type of cell of CNS is likely to clear necrotic tissue in the affected area? → answer = microglial cells (macrophages of the CNS).
- What cell does HIV infect in the CNS? → answer = microglia; microglia in the CNS are one of the major cellular reservoirs of HIV.
- 23F + Hx of migraines + Q asks which chemical mediator may be contributor to her Sx; answer = substance P.
- 44M + long Hx of alcoholism + stumbles to the right when ambulating + Romberg sign intact; where is the location of his pathology? → answer = right side of cerebellum; cerebellar lesions always ipsilateral; lateral aspects of cerebellum result in ipsilateral limb ataxia; midline cerebellar lesions (vermis) result in truncal ataxia; Romberg sign is (+) if etiology for ataxia is dorsal columns (i.e., tabes dorsalis in neurosyphilis); Romberg (-) means cerebellar etiology on USMLE.
- 51F + irregular flailing movements of right side of body; Dx and site of lesion? → answer = hemiballismus; lesion of contralateral (left) subthalamic nucleus.
- 5F + vomiting recently + bilateral papilledema + impaired upward gaze + walks with shuffling gait + CT shows enlarged lateral and 3rd ventricles with 2 cm mass; Q asks most likely location of the mass; answer = pineal gland; Dx = Parinaud syndrome (upward gaze palsy secondary to pinealoma); bottom line: brain tumor in kid + can't look up = pinealoma; vomiting usually in the morning (sign of increased intracranial pressure).
- 7M + ingested substance under the kitchen since + vomiting; what part of the brain is most likely responsible for the vomiting, and where is it located? → answer = area postrema, which is a chemotactic trigger zone for emesis; it is located on the dorsal aspect of the medulla at the caudal 4th ventricle.



- 51M + hereditary hemochromatosis + isolated decrease in serum gonadotropins; where is the most likely location of the pathology? → answer = hypothalamus → iron deposition in hypothalamus resulting in decreased GnRH; if deposition were in anterior pituitary, then we'd likely get a decrease in the other anterior pituitary hormones as well, rather than an isolated decrease in LH and FSH.
- What are the hormones secreted by the anterior pituitary? → LH, FSH, TSH, ACTH, GH, prolactin.
- What are the hormones **stored by** the posterior pituitary? → vasopressin (ADH), oxytocin; these hormones are **not** synthesized in the posterior pituitary; they are synthesized in the hypothalamus and then merely stored in the posterior pituitary.
- What are the hormones secreted by the hypothalamus that regulate the anterior pituitary hormones? → GnRH (stimulates LH, FSH); TRH (stimulates TSH); CRH (stimulates ACTH); GHRH (stimulates GH); dopamine (**inhibits** prolactin).
- 16F + MVA + pituitary stalk is severed; which hormone will increase? → answer = prolactin; all hypothalamic hormones are stimulatory except for dopamine's actions on prolactin, which are inhibitory.
- 54F + decreased fluency of spontaneous speech after a cerebral infarction + comprehension is normal + examination shows weakness of lower 2/3 of right side of face; Dx? + which vessel did the stroke occur? → answer = Broca aphasia; left middle cerebral artery.
- 65M + BP 140/100 + 2/5 muscle strength in left leg + brisk reflexes in left leg; which vessel is the most likely site of the stroke? → answer = right anterior cerebral artery.
- 24F + complains that food has lost its flavor ever since she got hit in the head during a softball game last week; damage to which structure may have occurred? → answer = cribriform plate; olfactory nerve lesion leads to decreased perception of taste.
- 81F + difficulty fastening buttons + weakness of intrinsic muscles of hands + loss of sensation in 5th fingers; Q asks for Dx at level of spine → answer = C7-T1 foraminal stenosis.
- 34F + Hx of MS + two-week Hx of severe shooting pain in left cheek brought on by brushing teeth; what is the most likely location of the demyelination leading to this patient's presentation? → answer = pons (giving rise to trigeminal nerve; in this case MS patient with trigeminal neuralgia-like presentation from central lesion).

- 61F + brought to ER by police after she was wandering around the city confused + disheveled + broad-based gait and nystagmus + results of alcohol and drug screening are negative; brain MRI will show atrophy of what? → answer = mammillary bodies; Dx is Wernicke encephalopathy (ACOW → Ataxia, Confusion, Ophthalmoplegia, Wernicke encephalopathy).
- 55M + two-week Hx of constant and intense pain of left arm + Hx of stroke 5 months ago resulting in loss of sensation to left side of body + examination shows decreased light touch, vibration, pain, and temperature over left side of body; Q asks: cause of this patient's pain is most likely due to damage to where? → answer = right thalamus; thalamic pain syndrome is a HY post-stroke cause of pain.
- 42F + struck in left eye by a golf ball + presents with diplopia, periorbital swelling, and enophthalmos + upward gaze is impaired + sensation is decreased over the left zygoma + x-ray shows orbital floor fracture; entrapment of which two muscles best explains this patient's presentation? → answer = inferior rectus and inferior oblique.
- 77M + sudden-onset miosis of left eye + ptosis of left eyelid + dysphagia + difficulty speaking + pain and Temperature are decreased on left side of her face and right side of body; which artery is most likely occluded? → answer = left posterior inferior cerebellar artery; Horner syndrome as associated with PICA infarct (lateral medullary syndrome) is HY.
- 15M + pain and tingling in her right hand when she goes backpacking + symptoms resolve five minutes after she takes her backpack off; Dx? → thoracic outlet syndrome due to first cervical rib.
- 26F + tonic-clonic seizure + immediately prior smelled foul odor for several seconds; where did the seizure most likely originate from? → answer = temporal lobe.
- 69M + stroke + hoarse voice + can't detect pinprick or cold on right side of face or left side of body; what's the location of the stroke? → answer = lateral medulla (right PICA); once again, lateral medullary syndrome = **PICAchew** (**Pikachu**) → **dysphagia**/hoarseness +/- Horner syndrome.
- 18M + used "synthetic heroine" at party + tight muscles + shuffling gait + bradykinesia + afebrile and pupils normal; Q asks location of the brain affected? → answer = substantia nigra; MPTP-induced Parkinsonism is assessed on both Step 1 and 2CK; MPTP is sold as "synthetic heroine" and can cause Parkinsonism.
- 28F + urinary incontinence + blurry vision + diminished vibration in the toes + unable to maintain balance with eyes closed; Q asks damage to which spinal tract? → answer = fasciculus gracilis of the

dorsal columns; in contrast, fasciculus cuneatus would do vibration/proprioception of upper limbs; diagnosis here is MS presenting with classic urge incontinence, optic neuritis, and a (+) Romberg sign.

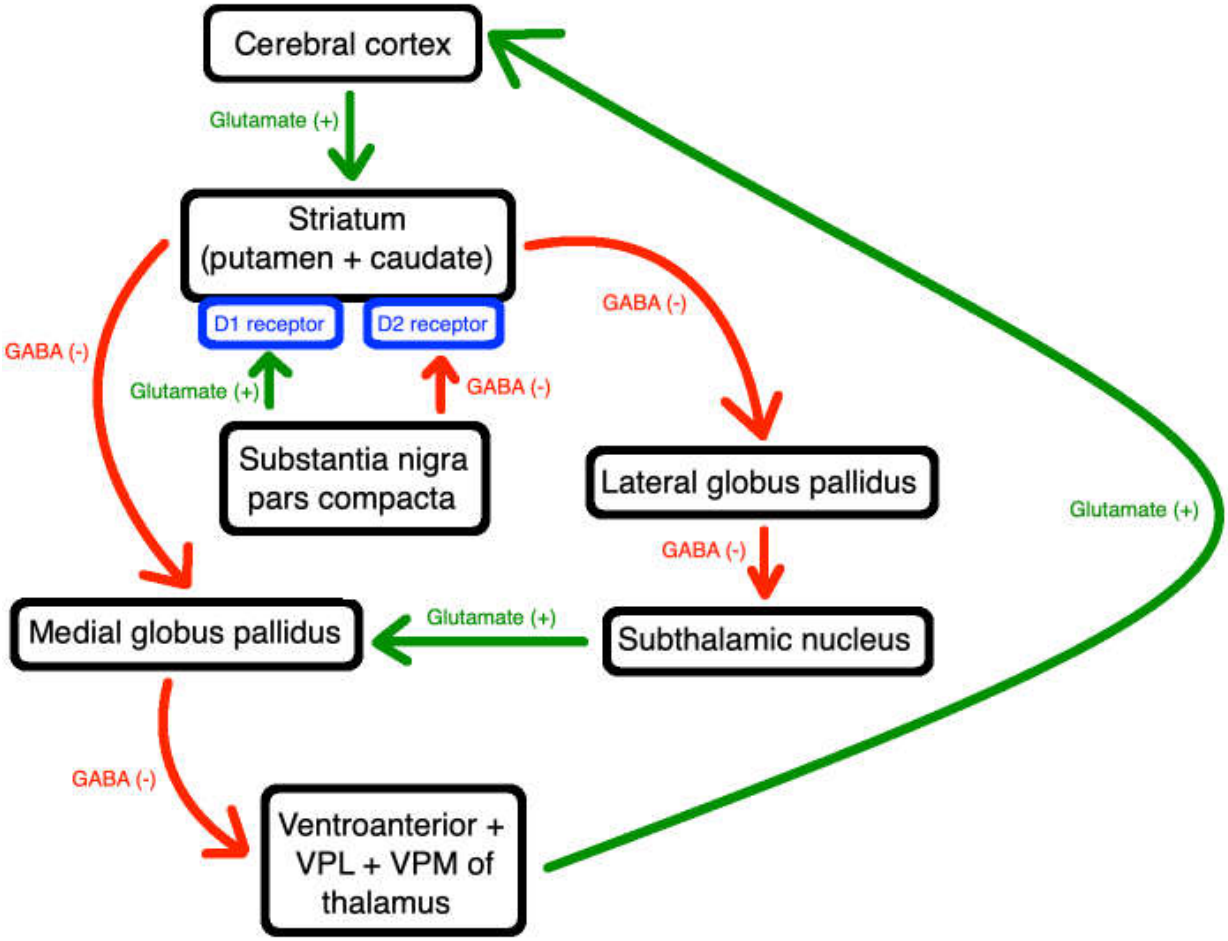
- 65M + Hx of T2DM, HTN, and hyperlipidemia + new-onset quadriplegia + impaired horizontal eye movements but vertical eye movements are intact + speech is dysarthric; Q asks for the part of the brain affected by the stroke; answer = pons; locked-in syndrome is caused by basilar artery stroke, resulting in classic triad of quadriplegia + aphonia + impaired horizontal eye movements.
- 3M + moderate strabismus; if not treated, she'll have deficits in depth perception due to lack of appropriate competitive interactions in the visual cortex; calcium entry through which receptor mediates the outcome of this competitive process? Answer = NMDA (glutamate) receptors; yes, weird, but handle it.
- Newborn + strabismus; what's the rationale for surgically correcting this problem during early childhood? → answer = if not, normal binocular vision won't develop.
- 23M + inferonasal "bleeding" of the pupil + incomplete closure of embryonic fissure of what structure in the right eye is most likely responsible?



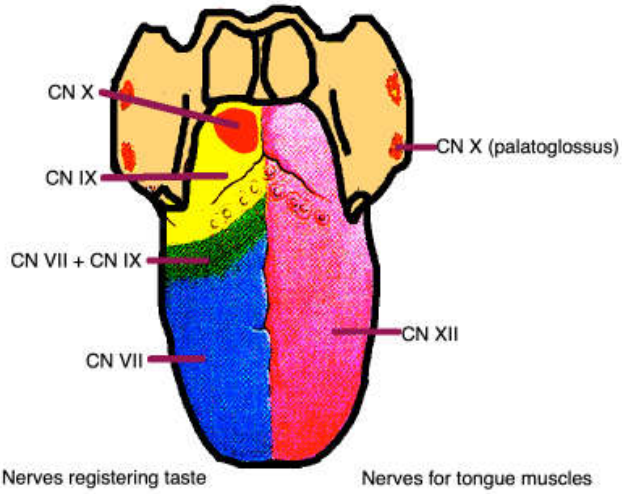
  - Answer = iris. Student says "wtf?" → CHARGE syndrome presents with coloboma of the eye, where the iris fails to develop properly, resulting in a "bleeding" of the pupil.
- 57F + invasive gall bladder cancer + 3-week Hx of severe left-sided back and abdominal pain; an operation scheduled to relieve pain would target which spinal tract? → answer = right spinothalamic tract (contralateral).
- 22F + fluoxetine prescribed; benefit occurs due to effects on neurons located where? → answer = raphe nuclei (serotonin-secreting neurons).

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Brain regions with high concentrations of certain neurotransmitter type	
Raphe nucleus	Serotonin
Nucleus basalis of Maynert (basal forebrain)	Acetylcholine
Locus coeruleus (pons)	Norepinephrine
Substantia nigra (midbrain)	Dopamine
Tuberomammillary nucleus (posterior hypothalamus)	Histamine
Cerebellum	GABA

- 42F + 4-month Hx of abnormal tingling that started in right hand has progressed to right arm and right side of face; cause of her Sx is a lesion located in which cerebral cortex gyri? → answer = left post-central (i.e., posterior to the central sulcus, which is where the primary sensory cortex is).
- 44M + progressive ataxia with unsteady gait + left-sided intention tremors; cause is damage to which brain area? → answer = left cerebellar hemisphere (cerebellar lesions lead to ipsilateral neurologic signs).
- 72F + loss of pain and temperature on left side of face and right of the body + paralysis of vocal cords on left + loss of gag reflex on left; what part of the brainstem is most likely affected? → answer = left dorsolateral medulla; lateral medullary syndrome; CN IX/V dysfunction; vocal cords are recurrent laryngeal nerves (of vagus); gag reflex is CN IX afferent and CN X efferent.
- 54F + 4-day Hx of hoarseness and difficulty swallowing + unable to elevate left side of palate; which CN is damaged? → answer = left vagus nerve; failure of elevation of ipsilateral palate causes contralateral uvula deviation.
- 54M + Hx of HTN + unable to recognize people's face + unable to recognize objects unless he touches them or hears sounds they make; damaged area of brain is supplied by what artery? → answer = posterior cerebral artery; prosopagnosia (inability to recognize faces) and visual agnosia (inability to recognize objects upon seeing them) are associated with occipital lobe lesions (PCA infarct).
- 61F + undergoes repair of leaking saccular aneurysm + postoperative CT scan shows widening of subarachnoid space; what changes in CSF are the most likely cause of these findings? → answer = decreased movement of CSF through the arachnoid villi.
- 24M + decreased taste to anterior 2/3 of left side of tongue; which nerve is likely damaged? → answer = left facial nerve.
- The thalamus notably does not receive sensory input from which nerve? → answer = olfactory nerve.
- Stroke affecting ventral posterolateral (VPL) & ventral posteromedial (VPM) nuclei of thalamus results in what kind of deficit? → pure sensory; VPL senses from body; VPM from face.
- 59M + Parkinson disease; loss of patient's substantia nigra is most likely to result in disinhibition in which brain area? → answer = striatum.



- 45M + uvula deviation to the right + dysphagia; which tongue muscle might be affected? → answer = palatoglossus (innervated by CN X); other tongue muscles innervated by CN XII.



- 29M + MVA + hematoma over mastoid processes bilaterally + rhinorrhea + otorrhea + raccoon eyes; Dx? → answer = base of skull fracture (classic tetrad).

- 31F + MVA + GCS of 6 + BP 165/90 + RR 10 + HR 55; what is the mechanism for this patient's hypertension? → answer = increased intracranial pressure (Cushing triad of HTN, bradycardia, bradypnea).
- 59F + metastatic breast cancer + absent corneal reflex on right; metastases to which cranial nerve exit point might be responsible for this neuro exam finding? → answer = superior orbital fissure → exit point for ophthalmic branch of trigeminal nerve; V1 → **S**tanding **R**oom **O**nly → **S**uperior orbital fissure, foramen **R**otundum, foramen **O**vale; for V1, V2, V3, respectively.
- 24M + herpes encephalitis + treated with IV acyclovir + months later has hyperphagia, hypersexuality, hyperorality; Dx? → answer = Kluver-Bucy syndrome (bilateral amygdala lesions); KB syndrome can be a rare complication of HSV encephalitis.
- 71F + stroke + now has agraphia, acalculia, finger agnosia, and left-right agnosia; what part of the brain did the stroke occur? → answer = angular gyrus (parietal lobe); Dx = Gerstmann syndrome.
- 23M + near-drowning episode with hypoxia to brain; what change might be seen in the hippocampi of this patient? → answer = "red neurons" → pathologic staining pattern of neurons during acute hypoxic injury; hippocampi notably susceptible to ischemia → results in anterograde amnesia.
- 40F + episodes of feeling like the room is spinning + sometimes associated with vomiting + physician is able to reproduce the Sx with the Dix-Hallpike maneuver; what is the mechanism for this patient's condition? → answer = "posterior semicircular canal otolith"; Dx = benign paroxysmal positional vertigo (BPPV); Tx with Epley maneuver.
- 40F + viral infection + tinnitus + hearing loss + vertigo; Dx? → answer = labyrinthitis.
- 40F + viral infection + vertigo; Dx? → vestibular neuritis.
- 31M + warm water instilled into left ear causes nystagmus, but warm water in right ear does not cause nystagmus; where is the site of the lesion? → right-sided vestibular apparatus.
    - Called the vestibular caloric reflex test.
    - Warm water instilled into ear causes endolymph on ipsilateral side to move, mimicking a head turn to the ipsilateral side. This causes eye movement to the contralateral side, with nystagmus back toward the ipsilateral side.
    - Cold water has the opposite effect. This mimics a head turn to the contralateral side, with the eyes moving toward the ipsilateral side, with nystagmus toward the contralateral side.

  - If nystagmus fails to occur with the side that is stimulated (i.e., the side that's warmer), then there is weakness of the vestibular apparatus on that side.

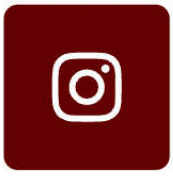
- Shining light in either eyes causes direct and consensual pupillary constriction; this clinical maneuver is a test of functional integrity of what? → answer = midbrain (Edinger-Westphal nucleus).
- 24M + left eye fails to constrict when light is shone in both the left and right eyes; where is the lesion → answer = oculomotor nerve (CN III).
- 24M + light directed into right eye results in constriction of right pupil but not left pupil + light directed into patient's left eye results in constriction of right pupil but not left pupil; where is the site of the lesion → answer = left oculomotor nerve (CN III); same thing as prior Q, but worded differently.
- 69M + smoker for 50 years + constricted left pupil + left eyelid droopy + skin dry on left side of face; where is the lesion? → Pancoast tumor of lung resulting in impingement on superior cervical ganglia (Horner syndrome).
- 24M + burning vesicular lesions around the lips; Q asks about the mechanism of this pathogen's presentation → NBME answer = "reactivation in sensory neurons" → herpes labialis (HSV1/2) → virus lies dormant in trigeminal nerve (V2 – maxillary branch; V3 – mandibular branch).
- 3-day-old male + born at home + umbilical cord tied with twine + arching of back; mechanism for disease process? → answer = "blockade of inhibitory neurotransmitter release" → tetanus toxin causes decreased presynaptic release of GABA and glycine by inhibiting SNARE protein; USMLE might show you a drawing of where the toxin acts, and the answer is inside the presynaptic terminal.

Rinne + Weber tests	
Rinne test	- A vibrating tuning fork is placed against the mastoid bone (bone conduction; BC). Once the patient can no longer hear it, it is removed from the bone and placed in front of the ear canal (air conduction; AC). - For both normal hearing and sensorineural hearing loss, AC > BC. - For conductive hearing loss, BC > AC.
Weber test	- A vibrating tuning fork is placed on the center of the forehead or top of the head, and the patient is asked if they hear the sound equally in both ears or if it's louder in one ear. - If it is louder in one ear, this is called "lateralization" to that ear.
Normal hearing	- Rinne shows air > bone conduction; Weber does not lateralize.
Conductive on left	- Rinne shows bone > air conduction; Weber lateralizes (is louder) to left.
Conductive on right	- Rinne shows bone > air conduction; Weber lateralizes to right.
Sensorineural on left	- Rinne shows air > bone conduction; Weber lateralizes to right.
Sensorineural on right	- Rinne shows air > bone conduction; Weber lateralizes to left.



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